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Simulations of NV color centers dynamics using QuTiP

São Carlos

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Simulations of NV color centers dynamics using QuTiP

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Advisor: Prof. Dr. Sérgio Ricardo Muniz

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*This work is dedicated to my lovely partner, Beatriz,
to the memory of my mother, Adriana,
and to my father, Lairson.*

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*“It’s the questions we can’t answer that teach us the most. They teach us how to think.
If you give a man an answer, all he gains is a little fact.”
Kvothe, The Wise Man’s Fear, Patrick Rothfuss*

ABSTRACT

DOMINGUES, A. V. **Simulations of NV color centers dynamics using QuTiP.** 2025. 79p. Dissertation (Master in Science) - Instituto de Física de São Carlos, Universidade de São Paulo, São Carlos, 2025.

Nitrogen-vacancy (NV) centers in diamond are a solid-state platform for quantum sensing¹ and information processing due to their spin-dependent optical transitions at room temperature². Accurate characterization of NV spin dynamics, including hyperfine interactions with nuclear spins, is essential both for optimizing sensing performance and for designing high-fidelity quantum controls. In this work, we use the open-source package/library Quantum Toolbox in Python (QuTiP)^{3, 4} to simulate two readout protocols (continuous-wave ODMR, pulsed ODMR) with and without the hyperfine coupling. By comparing simulated resonance positions and population dynamics against experimental and simulation profiles from the literature, we demonstrate that our QuTiP model reproduces the characteristic profiles present in other works. We analyze the computational cost of including hyperfine terms to the model via execution time averaging and show that built-in parallelization accelerates the no-hyperfine model but provides small returns when the coupling is included. Finally, we illustrate how simulation parameters align with predictions of DC magnetometry⁵. We also provide some ideas for extending this framework to GPU-accelerated and/or machine learning assisted parameter estimation and calibration^{6, 7}. Our soon-to-be open-source code⁸ provides a flexible foundation for both quantum sensing characterization and the development of NV-based quantum information protocols.

Keywords: NV center. Quantum control. Quantum simulation. QuTiP. Quantum information processing.

RESUMO

DOMINGUES, A. V. **Simulações da dinâmica de centros NV em diamante usando QuTiP**. 2025. 79p. Dissertação (Mestrado em Ciências) - Instituto de Física de São Carlos, Universidade de São Paulo, São Carlos, 2025.

Centros de vacância de nitrogênio (NV) em diamante são uma plataforma de estado sólido para sensoriamento quântico ¹ e processamento de informação quântica devido às suas transições ópticas dependentes de spin em temperatura ambiente ². A caracterização precisa da dinâmica de spin dos NV, incluindo interações hiperfinas com spins nucleares, é essencial tanto para otimizar o desempenho em sensoriamento quanto para projetar controles quânticos de alta fidelidade. Neste trabalho, utilizamos o pacote/biblioteca de código aberto *Quantum Toolbox in Python* (QuTiP)^{3, 4} para simular dois protocolos de medida (ODMR de onda contínua, ODMR pulsado) com e sem o acoplamento hiperfino. Ao comparar posições de ressonância via simulações e dinâmicas de população com os perfis experimentais e de simulação da literatura, demonstramos que nosso modelo em QuTiP reproduz os perfis característicos reportados em trabalhos anteriores. Analisamos o custo computacional de incluir termos hiperfinos no modelo por meio de médias de tempo de execução e mostramos que a paralelização nativa do QuTiP acelera o modelo sem interação hiperfina, mas oferece retornos modestos quando o acoplamento está presente. Por fim, ilustramos como os parâmetros de simulação reproduzem previsões de magnetometria DC ⁵. Também apresentamos ideias para estender esta plataforma para estimativas e calibrações de parâmetros acelerados por GPU e/ou assistidas por aprendizado de máquina ^{6, 7}. Nosso código, em breve *open-source* ⁸, fornece uma base flexível tanto para a caracterização de sensoriamento quântico quanto para o desenvolvimento de protocolos de informação quântica baseados em NV.

Palavras-chave: Centros NV. Controle quântico. Simulação quântica. QuTiP. Processamento de informação quântica.

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LIST OF ABBREVIATIONS AND ACRONYMS

CAPES	Coordenação de Aperfeiçoamento de Pessoal de Nível Superior
CW-ODMR	Continuous wave ODMR
CVD	Chemical vapor deposition
DND	Detonation nanodiamond
dip sep	dip separation
ES	Excited State
GKSL	Gorini-Kossakowski-Sudarshan-Lindblad
GPU	Graphics processing unit
GS	Ground State
HF	Hyperfine
HPHT	High-pressure high-temperature
IFSC	Instituto de Física de São Carlos
IR	Infrared
ISC	Intersystem crossing
LAC	Level anticrossing
MO	Molecular Orbital
MS	Metastable state
MW	Microwave
NV	Nitrogen-Vacancy
Nvidia CUDA-Q	Nvidia Compute Unified Device Architecture for Quantum
ODMR	Optically Detected Magnetic Resonance
PL	Photoluminescence
Pul-ODMR	Pulsed ODMR
QIP	Quantum information processing

Qiskit	Quantum Information Software Kit (by IBM)
QuTiP	Quantum Toolbox in Python
QuTiP-JAX	JAX backend for QuTiP
QuTiP-QOC	QuTiP-Quantum Optimal Control module
USP	Universidade de São Paulo
V	Vacancy
ZFS	Zero-Field Splitting
ZPL	Zero phonon line

LIST OF SYMBOLS

$\in$	is an element of
$\ \cdot\ $	Usual Euclidean norm in $\mathbb{R}^3$
$ \cdot $	Usual norm in $\mathbb{R}$
NV^0	Nitrogen vacancy in charge state 0
NV^+	Nitrogen vacancy in charge state +1
NV^-	Nitrogen vacancy in charge state -1 , also referred as NV
^{14}N	Nitrogen 14 isotope
^{15}N	Nitrogen 15 isotope
^{13}C	Carbon 13 isotope
Ω_R	Rabi frequency in MHz
W_P	Optical pumping rate (laser excitation rate)
k_{ij}	Transition rate between states i and j
n_i	Projection operator onto state i , <i>i.e.</i> , $ i\rangle\langle i $, also referred to as, population/polarization of state i
Δn_i	Absolute difference of polarization of state i between models, <i>i.e.</i> , $\Delta n_i = n_i^{(\text{ModelX})} - n_i^{(\text{ModelY})} $
n_{ES}	Sum of the polarizations of ES
$ i\rangle$	Ket state i , <i>i.e.</i> , $ i\rangle$
$\vec{B}_0$	Static magnetic field vector
B_0	$\ \vec{B}_0\ $ in G
θ_B	Polar angle of magnetic field
φ_B	Azimuthal angle of magnetic field
θ_n	Polar angle of nearby nucleus
φ_n	Azimuthal angle of nearby nucleus

D_i	Zero field splitting of $i \in \{\text{GS}, \text{ES}\}$
E_i	Strain splitting of $i \in \{\text{GS}, \text{ES}\}$
T_1	Relaxation time
T_2	Dephasing time
T_2^*	Dephasing time due to ensemble interaction
S	Spin of NV
m_S	Spin number of the state of NV
I	Spin of a nucleus near NV
m_I	Spin number of a nucleus state near NV
H_j^{Mk}	Hamiltonian of model k , with $j \in \{\text{GS}, \text{ES}\}$
$H_{j,\text{HF}}^{Mk}$	Hyperfine interaction Hamiltonian of model k , with $j \in \{\text{GS}, \text{ES}\}$
$\vec{S}_j$	Spin operator vector of NV, with $j \in \{\text{GS}, \text{ES}\}$, <i>i.e.</i> , $\vec{S}_j = (S_{x,j}, S_{y,j}, S_{z,j})$, where $S_{x,\text{GS}} = \frac{ 3\rangle\langle 1 + 1\rangle\langle 3 + 1\rangle\langle 2 + 2\rangle\langle 1 }{\sqrt{2}}$, $S_{y,\text{GS}} = \frac{ 3\rangle\langle 1 - 1\rangle\langle 3 + 1\rangle\langle 2 - 2\rangle\langle 1 }{i\sqrt{2}}$, $S_{z,\text{GS}} = 3\rangle\langle 3 - 2\rangle\langle 2 $, $S_{x,\text{ES}} = \frac{ 6\rangle\langle 4 + 4\rangle\langle 6 + 4\rangle\langle 5 + 5\rangle\langle 4 }{\sqrt{2}}$, $S_{y,\text{ES}} = \frac{ 6\rangle\langle 4 - 4\rangle\langle 6 + 4\rangle\langle 5 - 5\rangle\langle 4 }{i\sqrt{2}}$ and $S_{z,\text{ES}} = 6\rangle\langle 6 - 5\rangle\langle 5 $

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1 INTRODUCTION

Nitrogen vacancy (NV) centers have been studied for their potential in quantum metrology since their optical properties were observed at room temperature with scanning confocal optical microscopy in 1997. Their successful use as a platform for quantum sensing started in 2008 with magnetic sensing and has since expanded, with applications in quantum computation also being proposed.^{2, 14}

Characterization of the properties of any quantum sensor or computer,¹ NV being a particular case, is an important step in the development of applications. Having a simulation of the behavior of these particular systems provides a basis for such characterization. Since NV centers have been studied for about 30 years, the fundamental models used to analyze and characterize them have been well established through *ab initio* calculations^{15, 16, 17}, based on molecular orbital (MO) descriptions. These define a level-structure description with a base Hamiltonian. As better characterizations are developed and new phenomena are discovered, this base model can be tweaked and adjusted to obtain a more detailed Hamiltonian or a Gorini-Kossakowski-Lindblad-Sudarshan (GKSL) master-equation description of the system.^{18, 16, 19, 20, 21, 14, 22, 23, 24, 15, 17}

This description can be simplified or altered depending on the characteristic of the system that is the focus of the study; *e.g.*: how temperature changes the energy-level structure^{25, 26}, the behaviors of charge-state transfers^{27, 28, 29, 30}, the effects of different nuclei on the dynamics^{20, 21, 30, 13, 31}, the optical transfer rates between states^{11, 9, 13}, or how magnetic^{32, 33, 34, 31} or electric fields^{5, 14} affect the system. Of course, these effects can be complementary within a single model, and the researcher can study more than one at a time. Inspired by the fact that there are many models and ways to tackle this problem and that few studies^{25, 20} try to unify and improve simulation approaches, this work uses a well-known open-source package, namely the Quantum Toolbox in Python (QuTiP)^{3, 35, 4}, to investigate how well it can handle the complexities of simulating NV-center physics. The goal is to make this implementation available as open-source code⁸ for the community to improve upon and to facilitate parts of the work involved in studying the versatile quantum platform that NV centers provide.

For that purpose, we start with an overview of the main characteristics of NV centers in chapter 2. In section 2.1, the physical properties that make them an interesting platform for quantum applications are described. In sections 2.2 and 2.3, there is a description of the optical and magnetic transitions that they can exhibit. In section 2.4, a collection of techniques for measuring and preparing states is presented, and in section 2.5, it is explained how NV centers can be used as a quantum sensing platform.

In chapter 3, we review the three models ^{11, 12, 13} from the literature that were used to build the simulations presented in this work. Since QuTiP was used for this purpose, we also provide a brief introduction to QuTiP and a simple description of how the simulations were structured.

In chapter 4, we present the main results of this work. Regarding the simulation results, they are organized around two aspects: the capability to reproduce results from the literature (either simulations or experimental data) and the runtime performance, depending on the physical effects included. Finally, in chapter 5 there is a summary of the findings and future steps in the development of this tool, along with suggestions for applications for which it can be useful, such as machine-learning-assisted control or characterization. ^{19, 36, 37, 38, 21, 6, 39, 40, 41, 18, 7}

2 NV CENTER IN DIAMOND

2.1 Physical description

Diamond is one of the 3D crystalline forms of carbon, the other being graphite, and is most commonly found in a strain-free tetrahedral array but also has a hexagonal crystallographic form. It can be synthesized via natural or artificial means; there are three artificial sources of diamonds; high-pressure high-temperature (HPHT), detonation nanodiamond (DND), and chemical vapor deposition (CVD), the last commonly used for high precision fabrication ^{I, 42, 43}

Diamonds have incredible physical and chemical properties, such as high electron and hole mobilities, high thermal conductivity, and others summarized in Table 1. Two of high interest are the optical — with nearly constant refractive index, dielectric constant, and the widest spectral transmission range of all known solid materials — and electronic band gap of ≈ 5.5 eV (225 nm, UV) ¹⁸ at 300 K, which makes them a wide band gap semiconductor when substitutional doping is made. Impurities and defects can change the optical properties of diamond, for example allowing one-phonon infrared absorption. More than 500 optically active defects in diamond have been reported ⁴².

Table 1 – Selected properties of diamond.

Properties/Materials	Diamond	Si	4H-SiC	GaN
Bandgap (eV)	5.47	1.12	3.26	3.5
Electron-mobility (cm^2/Vs) at 300 K	1900-2300	1500	900	1250
Hole mobility (cm^2/Vs) at 300 K	1500-2300	600	100	200
Dielectric constant	5.7	11.9	9.7	9.5
Thermal conductivity (W/cmK)	25	1.48	4.9	1.3
Electron-saturation velocity (1×10^7 cm/s)	2.7	1	2.7	2.7
Breakdown field (1×10^5 V/cm)	100	3	30	30
Debye temperature (K)	1860	645	1200	608
Hardness (kg/mm^2)	10,000	1000	4000	

Source: Adapted from NEBEL ².

One of these defects is the nitrogen vacancy (NV) center structured as a single nitrogen (N) substitutional atom with one vacancy (V) adjacent to it. N-type impurities can also occur in other ways, such as clusters (the most common form in natural diamonds) or as isolated substitutional nitrogen, known as P1 (or C) centers, which act as deep donors with an ionization energy of 1.7 eV. Other kinds of N centers can act as electron donors,

^I although HPHT is also used for the fabrication of NV centers, a more precise control fabrication is possible when using CVD, but centers fabricated using HPHT have higher T_1 (lattice relaxation) times

such as divacancies, NVH, NVN. Some other defects that can occur when fabricating NVs or other color centers^{44, 2, 45} — which can also be other types of diamond doping with single atomic centers — are silicon-vacancies (SiV), germanium vacancies (GeV), Ni, Cr. An extra thing that can be done is the incorporation of atoms with specific nuclear spins, willingly in the fabrication, as an enrichment of the sample, to control dephasing times and other properties. These are most of the possible options available for manufacturing and working with quantum computation, quantum information processing, or quantum sensing using diamond defects.^{2, 46, 47}

The diamond's crystalline quality influences the characteristics and sensitivities of single-center or ensembles of NV centers. The spin dephasing time (T_2 for single centers and T_2^* for ensembles), measurable via techniques like Rabi oscillations or Hahn echo experiments, is a parameter influenced by such properties. As an example, the large Debye temperature of diamond gives rise to negligible decoherence via phonon interaction, also the spin decoherence which, in NV centers ($S = 1$), primarily arises from interactions with nearby nuclear and paramagnetic spins, for example ^{13}C nuclei — with natural abundance at about 1.1% — can dominate this characteristic, such that, T_2 can be around $600\ \mu\text{s}$ at room temperature, while reducing the ^{13}C concentration to approximately 0.3% can extend T_2 to about 1.8 ms. It has been reported that naturally occurring NV centers in CVD diamond exhibit spin coherence times longer than centers produced via ion implantation and subsequent annealing. After the invention of NV-based magnetic sensing, the platform has been used to detect a range of physical quantities including magnetic moments, electric fields, charge, voltage, current, orientation, strain, temperature, and pressure^{2, 45}.

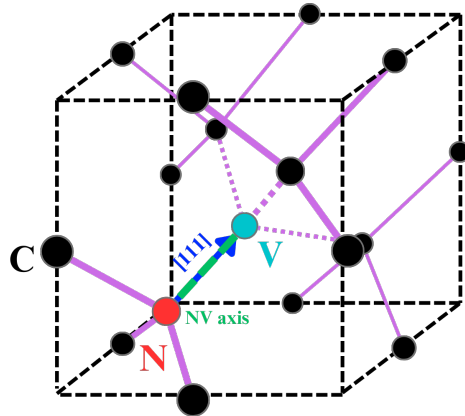


Figure 1 – Representation of the NV in the crystalline structure of the diamond. Here, the NV principal axis is aligned with the crystallographic $\langle 111 \rangle$ direction, consistent with its trigonal (C_{3v}) symmetry.

Source: Adapted from BARRY *et al*⁵.

NV centers can adopt one of the four possible crystallographic orientations. A representation of NV in the diamond lattice is shown in Fig. 1. It can be positively

(NV^+)^{48, 14}, neutral (NV^0) or negatively charged (NV^-). Most of the quantum applications of NV centers use the negatively charged state due to its favorable optical and spin properties that allow for sensing and quantum information processing (QIP)^{5, 47, 45, 14}; for this reason, unless specified otherwise, from now on “NV” will refer to NV^- .

Sometimes, due to optical excitation, NV^- can transition to NV^0 , temporarily or permanently², this phenomenon, which is well discussed and analyzed in some studies, can be used to improve the preparation or readout of states^{28, 49, 27, 50, 30, 29}. This work will not go into any detail about these mechanisms, although a description of them could be adapted for inclusion in our simulations in future studies.

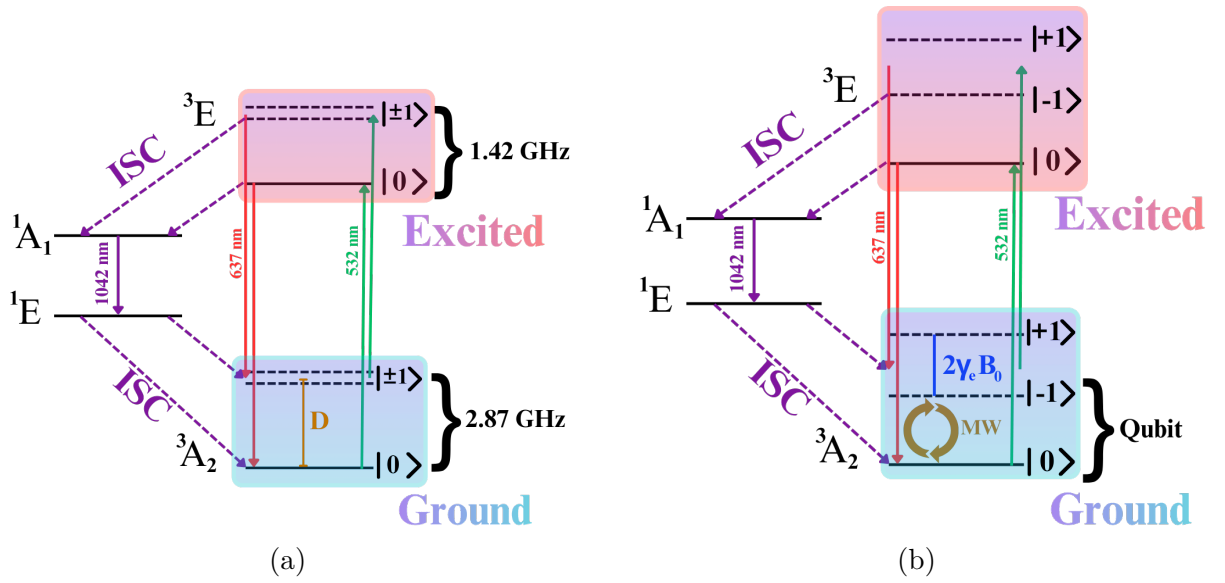


Figure 2 – NV center energy level diagrams comparing (a) optical transitions and energy level structure of the NV center at zero magnetic field with zero field splitting (light gold line) of the ground state. Optical transitions (solid arrows) and inter-system crossing (ISC) non-radiating pathways (dashed arrows) are shown, with no microwave transitions. (b) Energy level scheme under static magnetic field B_0 , showing Zeeman splitting (blue line), proportional to $\gamma_e B_0$, of spin sublevels ($m_s = -1, +1$) and microwave transitions (curved dark gold/ arrows) between ground state spin states, where γ_e is the electron gyromagnetic ratio. Distances of magnetic levels are not in scale to optical transition energies.

Source: By the author.

The NV center exhibits two triplet manifolds ($S = 1$), an optical ground-state (GS), which is a 3A_2 system, and an optical excited-state (ES), which being a 3E system. There is also an intermediate singlet manifold ($S = 0$) with metastable states (MS) 1A_1 and 1E ^{18, 12, 51, 5, 10}. The simplified Jablonski diagram of NV centers is shown in Fig. 2 and represents a simplified version of the NV energy states in room temperature. The low and intermediate temperature have different energy splitting and eigenstates and this phenomenon is explored in some works.^{25, 22, 12} The MS participate in a process called

intersystem crossing (ISC) which will be explained better in next section. Although we have different ways of describing these states and other possible transitions mechanisms, here we follow this description because it being the one that is used in many cases for a basis description of NV in literature^{18, 12, 2, 48, 5, 20, 21}.

The numbering convention shown in Fig. 3 will be used throughout the whole text from now on^{II}. If we need to talk about other possible processes that occur, and for that include extra energy levels, we will maintain this notation convention for the NV but include specific naming for the added energy levels.^{12, 18, 27, 25, 52, 53}

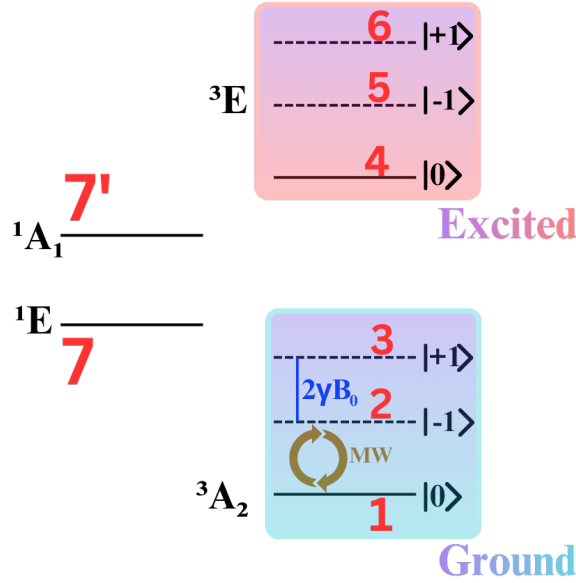


Figure 3 – Extended eight-state model differentiating singlet metastable states as (7/7'), built based on Figure 2b. State labeling: 1-3 (3A_2), 4-6 (3E), 7' (1A_1), 7 (1E).

Source: By the author.

2.2 Optical transitions

The optically active states in NVs are the charged states NV^- (Zero-phonon line (ZPL) around 637 nm) and NV^0 (ZPL around 575 nm), as previously stated. Light in a range of 480–640 nm can be used to optically pump NVs and promote a broad emission spectrum from 637 nm to 800 nm, a representation of this emission spectrum can be seen

^{II} Based on the energy scheme presented in Fig. 2, to simplify how to address each level, *i.e.*, remove the need to use phrases such as: "The optical ground state $m_S = 0$ transitions to the optical excited state $m_S = 0$ " (although being used some times when necessary); which could be called $|0\rangle_{GS}$ and $|0\rangle_{ES}$, but were not in the light of both simplicity and using notations close to those found in literature^{9, 11, 18}, we choose to name the $m_S = 0, -1$ and 1 states of the optical ground state as $|1\rangle$, $|2\rangle$ and $|3\rangle$ respectively, doing the same for the optical excited states but as $|4\rangle$, $|5\rangle$ and $|6\rangle$, and both metastable states as 7 and 7' (this allows to refer separately to the two levels when needed) using them as one state $|7\rangle$

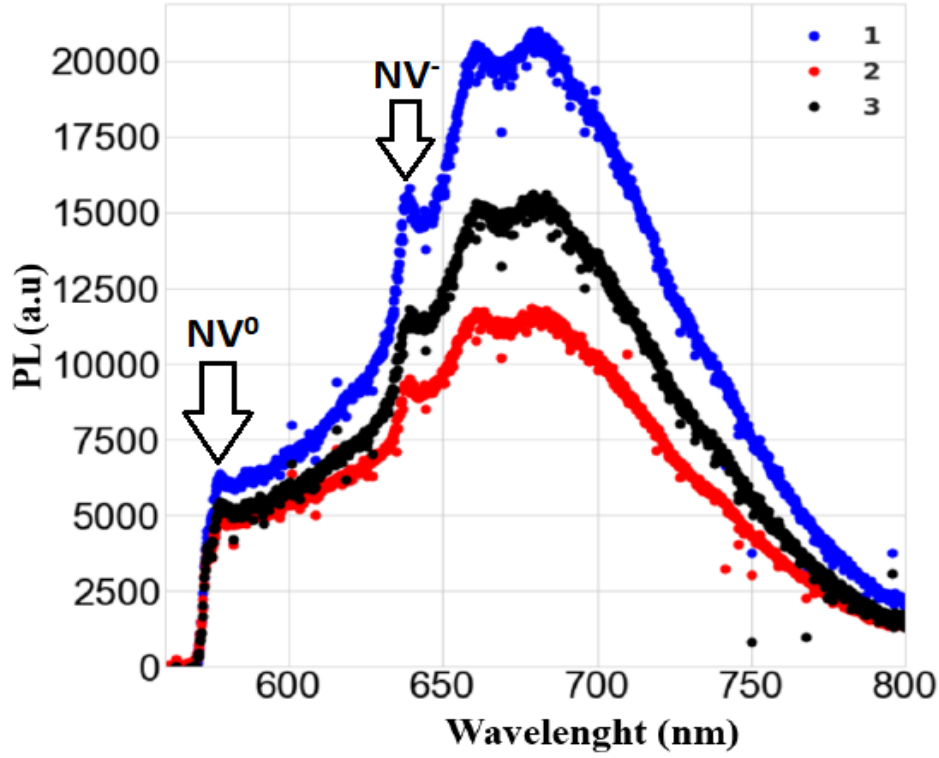


Figure 4 – The spectrum of emission of NV centers showing the ZPLs of NV^- and NV^0 taken from 3 different regions of a diamond sample

Source: Adapted from SEGURA ¹⁰.

in Fig. 4. At room temperature the ground states are naturally equally populated, a 532nm laser is most commonly used to promote the $|1\rangle \rightarrow |4\rangle$, $|2\rangle \rightarrow |5\rangle$ and $|3\rangle \rightarrow |6\rangle$ transitions. After optical excitation, there is either a direct decay back to the ground states emitting the photons in the spectrum range mentioned before — these transitions between 3A_2 and 3E are spin conserving and 3E have a lifetime of about 13 ns ² — or the ISC(called like this due to it being a non spin conserving process) decay path, *i.e.* $|4\rangle$ or $|5\rangle$ or $|6\rangle \rightarrow |7\rangle$, which the states 5 and 6 have more probability to go through than state 4, these transitions are non-radiative. Here, although treating the metastable states as one, it is worth mentioning that there occurs a decay from state $7'$ to 7 which emits a photon in the infrared (IR) with a ZPL at 1042nm, after that occurs the transitions $|7\rangle \rightarrow |1\rangle$ or $|2\rangle$ or $|3\rangle$, where there is preference of decay to state 1, these are also non-radiative transitions. This indicates that $|4\rangle \rightarrow |1\rangle$ and $|5\rangle$ or $|6\rangle \rightarrow |7\rangle \rightarrow |1\rangle$ are strong decay paths ($\sim 6\times$ more likely¹⁸) and $|5\rangle \rightarrow |2\rangle$, $|6\rangle \rightarrow |3\rangle$ and $|4\rangle \rightarrow |7\rangle \rightarrow |2\rangle$ or $|3\rangle$ are weak decay paths, this process produces an increase in the population of state $|1\rangle$ up to about 80% and is known as optical pumping (it can also appear in the literature as the spin polarization of state $m_S = 0$).

If one prepares the states $|2\rangle$ or $|3\rangle$, for example, using coherent microwave (MW)

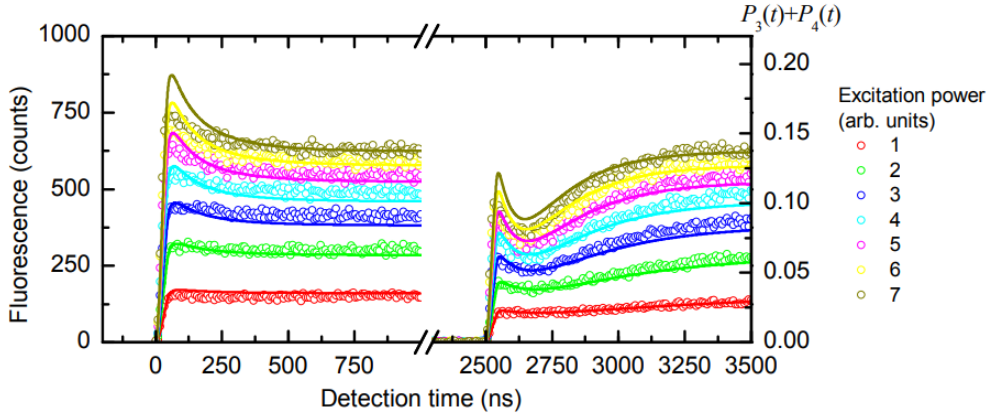


Figure 5 – NV photoluminescence (PL) versus detection time showing excitation-power dependence (experimental data ($\circ$) and model ($—$), both from Robledo *et al.*⁹). The first part corresponds to NV centers initialized in $m_S = 0$ and the second part to NV centers in $m_S = \pm 1$ after a MW π pulse. In our notation, the PL is proportional to the total excited-state population, $n_{\text{ES}}(t) = n_4(t) + n_5(t) + n_6(t)$. Excitation rates are integer multiples of 4 MHz.

Source: Adapted from ROBLED0 *et al.*⁹.

from the spin polarized state $|1\rangle$, these states will emit fewer photons (because of the effectively longer lifetime) when they are optically excited. The photoluminescence (PL) contrast between the state $|1\rangle$ and the state $|2\rangle$ or $|3\rangle$ could theoretically be up to about 30% (Fig. 5), but in practice it is usually much lower. This contrast enables photo detection and spin-state differentiation in the optical ground state.^{2, 54} State and Process Tomography can also be performed using this photo detection scheme.^{19, 38, 36, 55, 6}

Connecting theoretical simulations to experimental observables requires modeling the PL intensity. Although multiple PL models exist^{21, 25, 18}, we adopt the widely-used approximation found in Magaletti *et al.*¹¹, where, using our notation for the states, the PL intensity is proportional to the total excited-state population:

$$\text{PL}(t) \propto n_{\text{ES}}(t) = n_4(t) + n_5(t) + n_6(t), \quad (2.1)$$

with $n_i(t) = \text{Tr}\{|i\rangle\langle i| \rho(t)\}$.

To enable qualitative comparison between experimental data from the literature and simulation, we sometimes normalize the PL intensity when time is integrated, which is defined as

$$\text{PL}_{\text{norm}} = \frac{\int_{\tau_0}^{\tau_1} \text{PL}(t) dt}{\max_{t \in [\tau_0, \tau_1]} \text{PL}(t)}, \quad (2.2)$$

where τ_0, τ_1 represents the initial and final times chosen to integrate the photon count into the measurement duration, and the denominator corresponds to the maximum PL value within this integration time frame.

2.3 Magnetic transitions

In addition to optical transitions, there are transitions induced by magnetic fields in the form of microwaves (MW), which can be used to manipulate NV states, and radio-frequency (RF) to manipulate the states of nearby nuclei and consequently, via hyperfine interaction, manipulate the state of the NV centers.^{18, 41, 46, 56, 57, 40, 58, 47, 59}

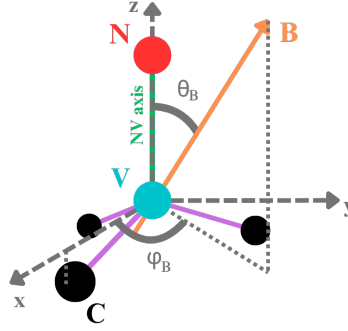


Figure 6 – Cartesian representation of the NV center under an applied magnetic field. The angles θ_B and φ_B are the polar and azimuthal angles of $\vec{B}$ (orange vector) with respect to the NV quantization axis.

Source: Adapted from ERNST²⁵.

In the presence of a static magnetic field, as shown in Fig. 6, the energy levels of the NV centers may shift because of the Zeeman interaction, so the manipulation using MW and RF must take this into account. In fact, even in the absence of field, both GS and ES have a zero-field splitting (ZFS) between $m_S = 0$ and $m_S = \pm 1$ with coefficients $D_{GS} \approx 2.87$ GHz and $D_{ES} \approx 1.42$ GHz, respectively.

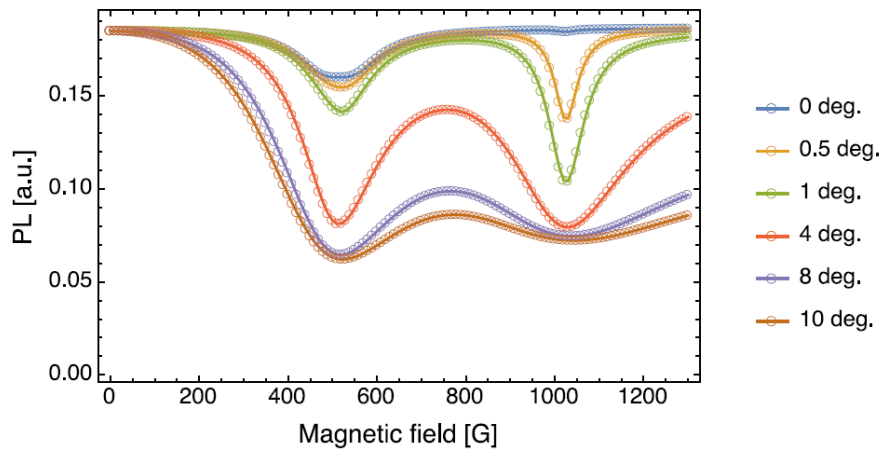


Figure 7 – Comparison of polar angles of the magnetic field θ_B for the PL intensity against magnetic field intensity.

Source: Adapted from HIROSE¹⁸.

However, when an external static magnetic field is aligned with the NV quantization axis (z-axis), *i.e.*, the axis connecting N and V, the degeneracy of $m_S = \pm 1$ is lifted for both ES and GS by a factor of $2\gamma_e B_z$, where the gyromagnetic ratio of the electron $\gamma_e = g_e \mu_B = 2.8 \text{ MHz/G}$ is composed of g_e , the g-factor of the electron, and μ_B , the Bohr magneton. When an off-axis magnetic field is introduced, the m_S states mix; this mixing becomes significant when $B_\perp$ is comparable to B_z , which may also affect the optical transitions (PL) ^{18, 41, 12, 60}, as shown in Fig. 7.

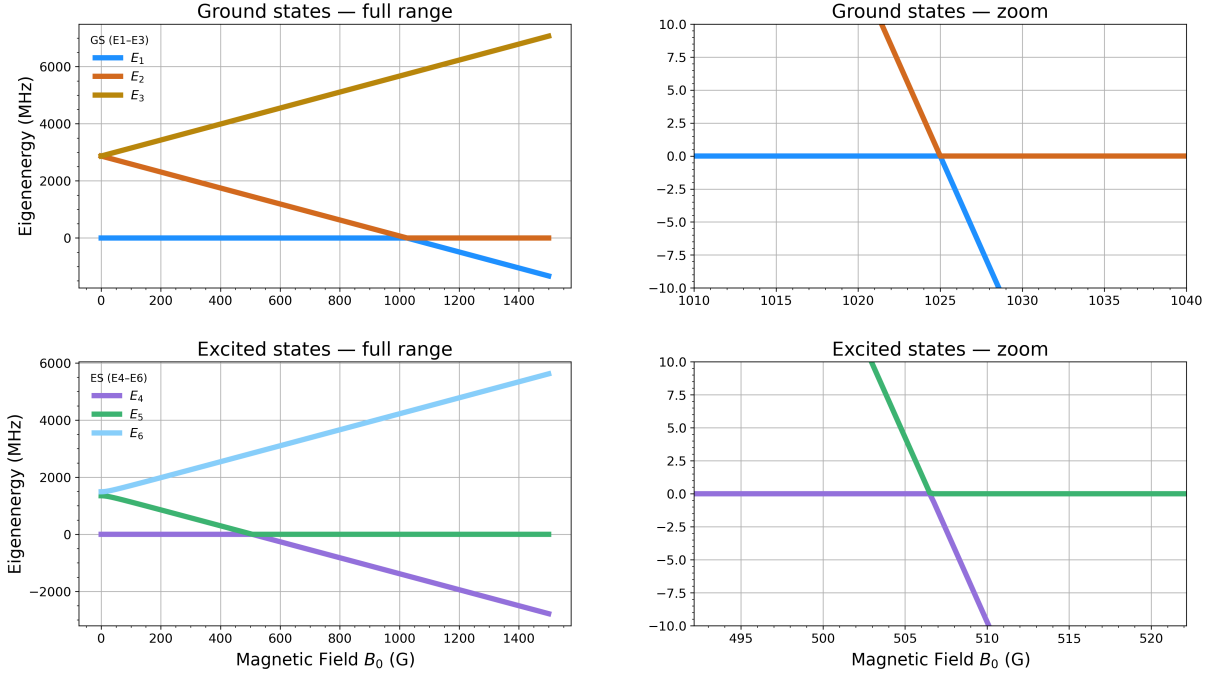


Figure 8 – Magnetic-field dependence of the NV energy levels for $\theta_B = 0$ and $\varphi_B = 0$. The left panels show the full field range for the ground- and excited-state manifolds, while the right panels show zoomed regions around the level anticrossings (LAC). The excited-state LAC occurs near $B_z \approx 508 \text{ G}$ and the ground-state LAC near $B_z \approx 1025 \text{ G}$.

Source: By the author.

Another effect that happens is the level anti-crossing (LAC) in the excited states (ESLAC), when $B_z > 508 \text{ G}$, and the ground states (GSLAC), when $B_z > 1025 \text{ G}$, with the mixture of states $m_S = 0$ and $m_S = -1$ when B_z is close either to 508 G, for the ES, or 1025 G, for the GS, this mixture of states is dependent of the spin (I) and state (m_I) of the nearby nuclei (*e.g.* for ^{15}N the mixture happens between $|m_S = 0\rangle |m_I = -1/2\rangle$ and $|m_S = -1\rangle |m_I = +1/2\rangle$ near GSLAC ³³ magnetic field value). The LAC puts the state $|5\rangle$, instead of being $|4\rangle$, as the lowest energy state for the ES (*i.e.*, the eigenstates relate to these levels swap) when past the threshold for ESLAC and $|2\rangle \leftrightarrow |1\rangle$ for the GS when past the threshold for GSLAC, these can be seen in Fig. 8 and Fig. 9, Tab. 2 and Tab. 3, where it is possible to see the energy level changes and inversion of the eigenstates related

Table 2 – Eigenstate compositions at $\theta_B = 0$, $\phi_B = 0$ (simplified model). The swap of the eigenstate identities $E_4 \leftrightarrow E_5$ between $B = 475$ G and $B = 508$ G highlights the excited-state LAC, while the inversion of E_1/E_2 beyond $B \approx 1025$ G corresponds to the ground-state LAC

B	E_1	E_2	E_4	E_5
0.0	$ 1\rangle$	$ 3\rangle$	$ 4\rangle$	$0.707 6\rangle + 0.707 5\rangle$
100.0	$ 1\rangle$	$ 2\rangle$	$ 4\rangle$	$0.993 5\rangle + 0.122 6\rangle$
350.0	$ 1\rangle$	$ 2\rangle$	$ 4\rangle$	$0.999 5\rangle$
475.0	$ 1\rangle$	$ 2\rangle$	$ 4\rangle$	$ 5\rangle$
508.0	$ 1\rangle$	$ 2\rangle$	$ 5\rangle$	$ 4\rangle$
525.0	$ 1\rangle$	$ 2\rangle$	$ 5\rangle$	$ 4\rangle$
750.0	$ 1\rangle$	$ 2\rangle$	$ 5\rangle$	$ 4\rangle$
1000.0	$ 1\rangle$	$ 2\rangle$	$ 5\rangle$	$ 4\rangle$
1025.0	$ 1\rangle$	$ 2\rangle$	$ 5\rangle$	$ 4\rangle$
1050.0	$ 2\rangle$	$ 1\rangle$	$ 5\rangle$	$ 4\rangle$
1200.0	$ 2\rangle$	$ 1\rangle$	$ 5\rangle$	$ 4\rangle$

Source: By the author.

to each energy level ^{III}. The transitions are done by choosing properly the MW or RF frequencies that are in resonance with the target states, where here multi-frequency pulses can be used, usually this is used for a more precise population of $|1\rangle$, or for aligning the magnetic field, since the PL changes a lot when the magnetic field is miss aligned from the z-axis by an angle (θ_B) even a little when near these regions, as is possible to see in Fig. 7. ^{33, 31, 34, 61, 62, 63, 37, 52}

Additionally, depending on the diamond sample, the GS and ES can also have a strain separation of $m_S = +1$ and $m_S = -1$, this effect acts as an intrinsic electric field and should be taken in account when measuring external electric fields and stress, although being observed while doing ODMR (a technique explained in Sec. 2.4) most of the times is a small effect so it can be ignored in most cases (regular values for these are $E_{GS} = 7$ MHz and $E_{ES} = 70$ MHz ^{32, 64, 65, 66, 67}), this effect is caused by mechanical stress on the diamond lattice, and it can be external (pressure) or internal (due to inhomogeneities in the lattice).

2.4 Spin dynamics: Controlling and reading the NV spin-qubits

Identifying the resonance frequencies of the NV centers is essential for controlling their states. Here we will be looking at three different readout sequences, two based on optically detected magnetic resonance (ODMR), which are only used for DC sensing, and one based on Ramsey interferometry, the three methods are useful for identifying the

^{III} These figures and tables were generated with Model 3 (with and without hyperfine coupling), as explained in the next chapter, and are included here solely for illustrative purposes. Be aware of eventual differences in notation, which are not relevant for what follows.

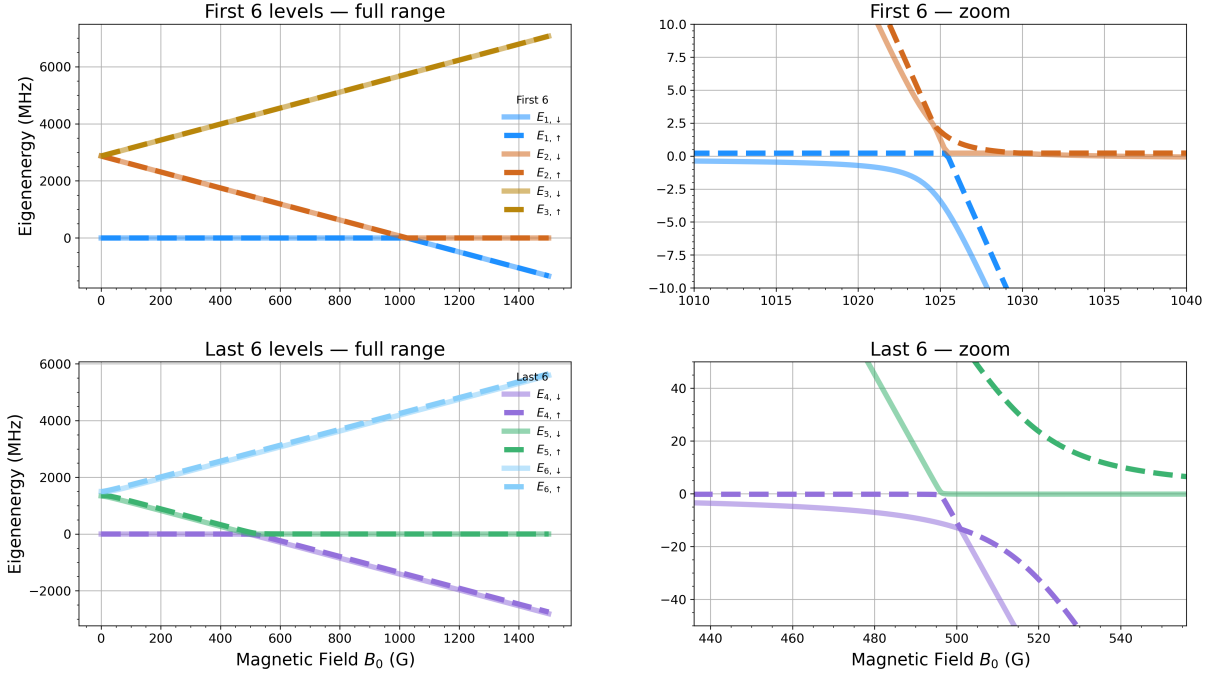


Figure 9 – Magnetic-field dependence of the NV energy levels including ^{15}N hyperfine interactions. The hyperfine coupling removes the crossing degeneracies and produces avoided crossings at the ESLAC (near $B_z \approx 508$ G) and the GSLAC (near $B_z \approx 1025$ G). Zoomed regions on the right highlight how the mixed eigenstates evolve across each anticrossing.

Source: By the author.

resonance frequencies and can also be used for sensing, *e.g.* magnetometry, where each of them have different sensitivities and ways to infer the magnetic field. The ODMR methods are based on the difference in the PL intensity of the NV center when the microwave field is in or off resonance with the NV center transitions, and the Ramsey method is based on the cumulative phase of a free precession time present in the protocol, this method can also be used to analyze T_2^* , and a modification to it can be used for analyzing T_2 .^{18, 41}

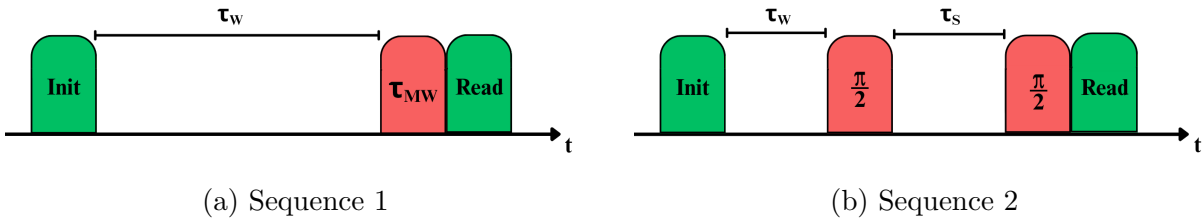


Figure 10 – Schematic timing diagrams of the simulated sequences/protocols (and the corresponding figures in Chapter 4), indicating laser (green), MW (red), and free-evolution intervals. (a) **Sequence 1**, in Magaletti *et al.*¹¹. (b) **Sequence 2**, for doing Ramsey magnetometry.

Source: By the author.

Table 3 – Eigenstate compositions at $\theta_B = 0$, $\phi_B = 0$ including ^{15}N hyperfine structure. The coefficients at $B = 508\text{ G}$ and $B = 1025\text{ G}$ show the partial mixing of the $m_S = 0$ and $m_S = -1$ subspaces at the excited- and ground-state LACs, respectively.

B	$E_{1,\downarrow}$	$E_{2,\downarrow}$	$E_{4,\downarrow}$	$E_{5,\downarrow}$
	$E_{1,\uparrow}$	$E_{2,\uparrow}$	$E_{4,\uparrow}$	$E_{5,\uparrow}$
0.0	$ 1, \downarrow\rangle$	$ 3, \uparrow\rangle$	$ 4, \downarrow\rangle$	$0.832 5, \uparrow\rangle + 0.555 6, \uparrow\rangle$
	$ 1, \uparrow\rangle$	$ 2, \downarrow\rangle$	$ 4, \uparrow\rangle$	$0.832 6, \downarrow\rangle + 0.555 5, \downarrow\rangle$
100.0	$ 1, \uparrow\rangle$	$ 2, \downarrow\rangle$	$ 4, \uparrow\rangle$	$0.994 5, \uparrow\rangle + 0.111 6, \uparrow\rangle$
	$ 1, \downarrow\rangle$	$ 2, \uparrow\rangle$	$ 4, \downarrow\rangle$	$0.990 5, \downarrow\rangle + 0.136 6, \downarrow\rangle$
475.0	$ 1, \uparrow\rangle$	$ 2, \downarrow\rangle$	$0.976 4, \uparrow\rangle + 0.219 5, \downarrow\rangle$	$ 5, \uparrow\rangle$
	$ 1, \downarrow\rangle$	$ 2, \uparrow\rangle$	$ 4, \downarrow\rangle$	$0.975 5, \downarrow\rangle + 0.219 4, \uparrow\rangle$
508.0	$ 1, \uparrow\rangle$	$ 2, \downarrow\rangle$	$0.999 5, \uparrow\rangle$	$ 4, \downarrow\rangle$
	$ 1, \downarrow\rangle$	$ 2, \uparrow\rangle$	$0.840 4, \uparrow\rangle + 0.543 5, \downarrow\rangle$	$0.840 5, \downarrow\rangle + 0.543 4, \uparrow\rangle$
525.0	$ 1, \uparrow\rangle$	$ 2, \downarrow\rangle$	$ 5, \uparrow\rangle$	$ 4, \downarrow\rangle$
	$ 1, \downarrow\rangle$	$ 2, \uparrow\rangle$	$0.830 5, \downarrow\rangle + 0.558 4, \uparrow\rangle$	$0.830 4, \uparrow\rangle + 0.558 5, \downarrow\rangle$
1000.0	$0.999 1, \uparrow\rangle$	$0.999 2, \downarrow\rangle$	$ 5, \uparrow\rangle$	$ 4, \downarrow\rangle$
	$ 1, \downarrow\rangle$	$ 2, \uparrow\rangle$	$ 5, \downarrow\rangle$	$ 4, \uparrow\rangle$
1025.0	$0.776 2, \downarrow\rangle + 0.631 1, \uparrow\rangle$	$ 2, \uparrow\rangle$	$ 5, \uparrow\rangle$	$ 4, \downarrow\rangle$
	$ 1, \downarrow\rangle$	$0.776 1, \uparrow\rangle + 0.631 2, \downarrow\rangle$	$ 5, \downarrow\rangle$	$ 4, \uparrow\rangle$
1050.0	$0.999 2, \downarrow\rangle$	$0.999 1, \uparrow\rangle$	$ 5, \uparrow\rangle$	$ 4, \downarrow\rangle$
	$ 2, \uparrow\rangle$	$ 1, \downarrow\rangle$	$ 5, \downarrow\rangle$	$ 4, \uparrow\rangle$

Source: By the author.

Ramsey interferometry sequence, illustrated in Fig. 10b, is done by applying a laser to the NV center for an initialization time (τ_I), to populate/polarize NV to the state $|1\rangle$, then waiting a time (τ_W) for the population from the $|7\rangle$ state to be fully transferred to the ground states finishing the optical pumping process, this is the full process of initialization of the system in state $|1\rangle$, after this a microwave field is applied for a time $\tau_{\pi/2} = \pi/2\Omega_R$ (Ω_R is the Rabi frequency) to create a superposition of $|1\rangle$ and one of the two $m_S = \pm 1$ GS either $|2\rangle$ or $|3\rangle$ (if no static magnetic field is present one gets a division of population between the states $|2\rangle$ and $|3\rangle$), the sequence continues with a free precession time τ_S , that accumulates a phase proportional to the static field being sensed ($\phi = \gamma_e B_0 \tau_S$), followed by applying a microwave field for a time $\tau_{\pi/2}$ to map the population difference between $|1\rangle$ and the other state chosen, the process finishes by the application of a laser for a readout time (τ_R) to measure the PL intensity which is integrated through a part or the whole τ_R . This process creates a mapping between τ and PL that is commonly called Ramsey fringes, the Fourier transform of it gives the detuning from the resonance frequency, while itself shows the loss of coherence due to interactions with the magnetic environment; this method can be used for both AC and DC sensing, where for AC some modifications need to be done in the protocol. ^{18, 54}

The continuous wave ODMR (CW-ODMR) is a simpler process, a laser and a microwave are applied together for a time τ_O and the PL intensity is integrated through τ_O , this process is done while the microwave field driving frequency is swept in a range of frequencies large enough to get the resonance frequencies of NV, the integrated PL intensity

against the swept frequencies plot will have a dip near the resonance frequencies, these dips can be approximated by Lorentzians or Generalized Gaussians, the separation between the dips is proportional to the static magnetic field being measured (dip sep $\sim 2\gamma_e B_0$), as can be seen by Fig 11. This process has a broadened linewidth of Lorentzians/Gaussians (~ 10 MHz), this happens because the measurement and GS excitation/oscillation happen at the same time, giving it a limited sensitivity.^{18, 54, 41, 5}

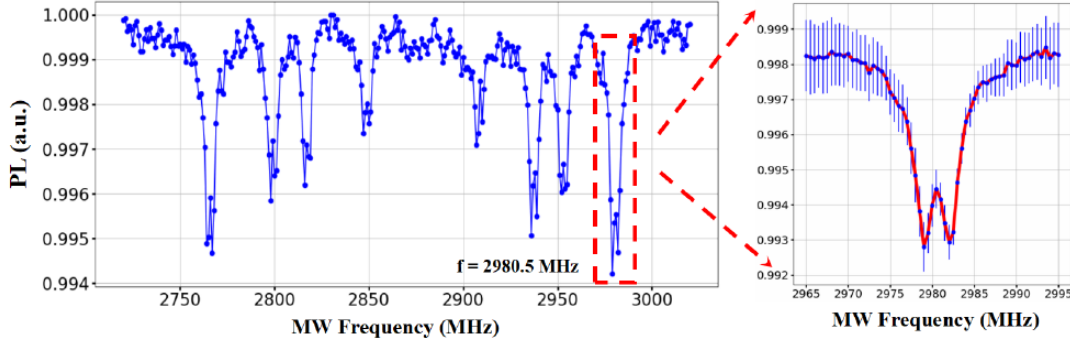


Figure 11 – Example CW-ODMR spectrum, from our lab¹⁰, of a diamond sample with an ensemble of NVs oriented along all four crystallographic directions (left). One resonance dip is enlarged on the right; the two small sub-dips arise from the ^{15}N hyperfine splitting. In this case, the applied static magnetic field is $B_z \approx 90$ G.

Source: Adapted from SEGURA¹⁰.

The pulsed ODMR is similar to the CW-ODMR in the sense that a swipe of the same range of driving frequencies is done and the PL over frequency profile has the dips that are proportional to the static magnetic field, but the difference lies in that the sequence happens similarly to the Ramsey method, to separate measurement from GS oscillations, which improve sensitivity, the Lorentzians/Gaussians have sharper linewidths (few hundreds of kHz). The sequences consist of an initialization laser time (τ_I), a polarization wait time (τ_W) ($|1\rangle$ initialization), followed by the application of a microwave field for a time, which to maximize sensitivity⁵⁴ must be the time of a π pulse ($|2\rangle$ or preparation $|3\rangle$), so at the same time as $\tau_\pi = \pi/\Omega_R$, finally a laser is applied for the readout time τ_R .^{54, 48, 5, 18, 41}

Zhang *et al.*⁵⁴ shows a few more methods and a comparison between sensitivities, Barry *et al.*⁵ goes in more detail about each protocol and sensitivities, Hirose¹⁸ and Schloss⁴¹ have many techniques on how to improve fidelity and sensitivity which are worth exploring, but here we focus on using only ODMR methods to show that we can reproduce experimental results with a simulation in QuTiP.

2.5 The NV qubit as a quantum sensor

Quantum sensing as described in Degen, Reinhard and Cappellaro¹ is the use of quantum objects or coherences to measure physical quantities (classical or quantum) or the use of quantum resources (*e.g.*, quantum entanglement) to improve measurement sensitivities or precisions beyond what is achievable classically. Quantum sensors must have some properties which are also stated in the work just mentioned, they must have discrete, resolvable energy levels that can be initialized in well known states and be readout, and also they should be coherently manipulated, at least in most of the protocols, and interact with the physical quantities aimed at being measured. NV centers surely have all these requirements and work in quantum sensing mostly in the coherently manipulated way just mentioned, they interact with magnetic and electric fields, pressure, temperature, nearby spins, and can indicate rotations in spatial directions, all properties that can be measured optically, with the initialization occurring optically as well.¹

A generic description of a quantum sensor involves the known characteristics of the sensor, the control fields, and the interaction of the sensor with the physical quantity being sensed. This can be put in a simple form as¹:

$$H(t) = H_0 + H_V(t) + H_{\text{control}}(t), \quad (2.3)$$

where H_0 encompasses the known characteristics of the system, $H_V(t)$ is its interaction with the sensed system/property, and $H_{\text{control}}(t)$ is the control interaction. The task of a quantum sensing experiment is choosing cleverly $H_{\text{control}}(t)$ to infer the characteristic of the signal $V(t)$. This task can be achieved in some different ways, but the usual procedure is to initialize the system in a known state, prepare a state, let the system evolve for a time t according to the propagator of Eq. 2.3 (i.e. $U_H(0, t)$), prepare the readout state, do the readout, and repeat to collect statistics for estimating the signal $V(t)$.¹

This procedure is very generic and can be optimized in many steps, including better characterization of the sensor, improvements in steps of the sensing protocols, and other possible nuances of each sensor or sensing scheme. So, automating some parts of these processes and having expected behaviors and characteristics depending on what is being done are good objectives to have in mind. Simulations can help with that, and for that, we need a theoretical description of the system and tools for simulating it and automating its control. Some theoretical descriptions of NVs and a possible tool for simulation and possibly automated control are presented in the next chapter.

3 MODELING OF THE PHYSICAL SYSTEM

Based on the description above, there are some models to describe the NV center, especially its interaction with nearby nuclei. Here we are working with three and they are derived from a base model, described by the spin Hamiltonian governing the energy structure of the nitrogen-vacancy (NV) center expressed as

$$H_j = D_j S_{z,j}^2 + \gamma_e \vec{B} \cdot \vec{S}_j + \gamma_n \vec{B} \cdot \vec{I} + Q I_z^2 + H_{j,\text{HF}}, \quad (3.1)$$

where $j \in \{\text{GS}, \text{ES}\}$ distinguishes between GS and ES. It is always good to keep in mind that the NV Hilbert space, in these cases, is a direct sum of the spaces of the ES, GS and MS, which in the latter case have dimensions 3,3,1, respectively¹⁹.

The first term, $D_j S_{z,j}^2$, represents the zero-field splitting (ZFS), with D_j as the ZFS constant. The following terms account for Zeeman interactions: $\gamma_e \vec{B} \cdot \vec{S}$ and $\gamma_n \vec{B} \cdot \vec{I}$ describe electron and nuclear spin couplings to the external magnetic field $\vec{B}$, respectively. Here, $\gamma_e = g_e \mu_B = 2.8 \text{ MHz/G}$ and $\gamma_n = g_I \mu_n$ (see Table 4) are the gyromagnetic ratios, while $\vec{S}$ and $\vec{I}$ denote the electron and nuclear spin operators, g_e and g_I are the g-factors for the electron and the relevant nucleus and μ_B and μ_n are the Bohr and nuclear magnetons.

The magnetic field $\vec{B}$ includes a static bias component given by

$$\vec{B}_0 = B_0 (\cos \varphi_B \sin \theta_B, \sin \varphi_B \sin \theta_B, \cos \theta_B), \quad (3.2)$$

which is depicted in Fig. 6, usually aligns with the NV quantization axis (z direction), and a time-dependent microwave component $\vec{B}_{\text{MW}}(t) = \sqrt{2} \Omega_R \hat{n} \cos(\omega t) / \gamma_e$. Here, instead of using the intensity of the MW field B_{MW} , which is proportional to the microwave power (P_{MW}), we used the Rabi frequency $\Omega_R = \gamma_e B_{\text{MW}}$, also $\hat{n}$ is the orientation of the MW field and ω is the driving frequency.

What happens regularly is that the microwave component is given transversely (x-y plane) to the quantization axis, so that the term $\vec{B}_{\text{MW}}(t) \cdot \vec{S}_j$ is given by

$$\vec{B}_{\text{MW}}(t) \cdot \vec{S}_j = \sqrt{2} \frac{\Omega_R}{\gamma_e} (S_{x,j} \cos(\phi_{\text{MW}}) + S_{y,j} \sin(\phi_{\text{MW}})) \cos(\omega t), \quad (3.3)$$

where ϕ_{MW} is the phase between the components of the microwave field, which is usually set to zero. The factor $\sqrt{2}$, guaranties that $S_{x,j}$ and $S_{y,j}$, turn into Pauli matrices $\sigma_{x,j}$ and $\sigma_{y,j}$, respectively, assuring that π -pulses happen with time π/Ω_R , without this $\sqrt{2}$ factor, π -pulses occur with a different time with respect to what they should be, this was done because usually the MW interaction is stated using Pauli matrices as $\Omega_R/2 \sigma_{k,j} \cos(\omega t)$, $k \in \{x, y\}$, and without the extra factor we are using spin matrices. The choice to model the MW field in this aspect is due to it inducing transitions between

$|0\rangle_{GS} \leftrightarrow |\pm 1\rangle_{GS}$ states, when $\omega \approx \omega_0 = D_{GS} \pm \gamma_e B_{0,z}$, and enabling qubit encoding. If a non-transverse MW field is applied, it creates level shifts that are time-dependent, and more complicated calibration protocols would be required to achieve the same results, if that is even possible. Maybe these properties can be useful for different types of application, but here we will not be exploring this.

When a static magnetic field is applied off-axis to the NV quantization axis, the resonance frequency changes and is given to the third order in B_0/D_{GS} , when $B_0 \ll D_{GS}$, by the equation ⁵

$$\begin{aligned} \omega_0 = D_{GS} \left[1 \pm \left(\gamma_e \frac{B_0}{D_{GS}} \right) \cos \theta_B + \frac{3}{2} \left(\gamma_e \frac{B_0}{D_{GS}} \right)^2 \sin^2 \theta_B \right. \\ \left. \pm \left(\gamma_e \frac{B_0}{D_{GS}} \right)^3 \left(\frac{1}{8} \sin^3 \theta_B \tan \theta_B - \frac{1}{2} \sin^2 \theta_B \cos \theta_B \right) \right], \end{aligned} \quad (3.4)$$

where $B_0 = \|\vec{B}_0\|$ and $\tan \theta_B = B_{0,z}/B_{0,\perp}$. For a more general form of Eq. 3.4, the resonance frequencies are found as $\omega_0(B_0, \theta_B) = E_{\pm}(B_0, \theta_B) - E_0(B_0, \theta_B)$, where $E_{\pm}, E_0$ are the solutions of the following equation ³⁴:

$$\lambda^3 - 2D_{GS}\lambda^2 + \left(D_{GS}^2 - (\gamma_e B_0)^2\right) \lambda + \frac{D_{GS}}{2} (\gamma_e B_0)^2 (1 - \cos(2\theta_B)) = 0. \quad (3.5)$$

The quadrupole term QI_z^2 is specific to ^{14}N ($Q = 4.96$ MHz), and arises from its $I = 1$ nuclear spin, unlike ^{15}N ($I = 1/2$). The final term, $H_{j,\text{HF}}$, represents the hyperfine interaction that connects the electron spin of the NV to nearby nuclear spins, such as the ^{14}N , ^{15}N , or carbon ^{13}C isotopes. Different models for $H_{j,\text{HF}}$ exist in the literature; here we will focus on two described in sub-sections 3.1.2 and 3.1.3. Before that, we will explain one model that has a good description for the optical excitations in sub-section 3.1.1; these optical excitations models don't vary too much, but they sometimes reflect the addition or exclusion of some transition paths.

Other terms can be added to the Hamiltonian relative to strain properties that are considered internal electric field effects, adding deviations in the electric field and stress measurements (this can also happen to magnetic fields smaller than the strain term) and inducing a small shift between $|\pm 1\rangle$ states, but this also leaves the scope of this work; for discussions on these terms, see Lee *et al.*⁵³, Barry *et al.*⁵ and Doherty *et al.*¹².

This formalism enables numerical simulation of population dynamics in NV centers while accounting for hyperfine interaction effects. Analytical solutions that reduce to rate equations exist only in the restricted case where MW fields and optical excitation are not simultaneously present ^{19, 21}. However, realistic readout protocols (discussed in 2.4) violate this condition, necessitating full quantum dynamical treatment for accurate simulation.

Table 4 – Key NV center constants for ^{14}N and ^{15}N . The quadrupole interaction Q is absent for ^{15}N . Hyperfine parameters $A_{\perp}$ and $A_{\parallel}$ differ between isotopes.

	^{14}N	^{15}N
D_{GS}	2.87 GHz	2.87 GHz
D_{ES}	1.42 GHz	1.42 GHz
γ_e	2.8 MHz/G	2.8 MHz/G
$A_{\perp}$	2.14 MHz	3.03 MHz
$A_{\parallel}$	2.70 MHz	3.65 MHz
Q	4.96 MHz	-
γ_n	0.308 kHz/G	0.432 kHz/G

Source: By the author.

3.1 Previous models in literature

3.1.1 Model 1: Magaletti *et al.*

This model uses the 7-level structure shown in Fig. 3. In the paper, the authors model the system through a set of rate equations as:

$$\begin{aligned}
\frac{dn_1}{dt} &= -n_1 W_P + n_4 k_{41} + n_7 k_{71} + \Omega_R \cdot n_c \\
\frac{dn_2}{dt} &= -n_2 W_P + n_5 k_{52} + n_7 k_{72} - \Omega_R \cdot n_c \\
\frac{dn_3}{dt} &= -n_3 W_P + n_6 k_{63} + n_7 k_{73} \\
\frac{dn_4}{dt} &= n_1 W_P - n_4 k_{41} - n_4 k_{47} \\
\frac{dn_5}{dt} &= n_2 W_P - n_5 k_{52} - n_5 k_{57} \\
\frac{dn_6}{dt} &= n_3 W_P - n_6 k_{63} - n_6 k_{67} \\
1 &= n_1 + n_2 + n_3 + n_4 + n_5 + n_6 + n_7 \\
\frac{dn_c}{dt} &= -\Gamma_2 \cdot n_c + \frac{\Omega_R}{2} \cdot (n_2 - n_1)
\end{aligned} \tag{3.6}$$

here, $n_i = \rho_{ii}$ is the population of level i of NV center and n_c is the coherence between level 1 and 2, with W_P being the laser pumping parameter, Ω_R the Rabi frequency, and Γ_2 the spin decoherence rate. We adapted this to a master equation using optical transitions modeled via Lindblad operators^{19, 11, 13}:

$$\begin{aligned}
L_{14} &= \sqrt{k_{14}} |4\rangle\langle 1|, & L_{25} &= \sqrt{k_{25}} |5\rangle\langle 2|, & L_{36} &= \sqrt{k_{36}} |6\rangle\langle 3|, \\
L_{41} &= \sqrt{k_{41}} |1\rangle\langle 4|, & L_{52} &= \sqrt{k_{52}} |2\rangle\langle 5|, & L_{63} &= \sqrt{k_{63}} |3\rangle\langle 6|, \\
L_{47} &= \sqrt{k_{47}} |7\rangle\langle 4|, & L_{57} &= \sqrt{k_{57}} |7\rangle\langle 5|, & L_{67} &= \sqrt{k_{67}} |7\rangle\langle 6|, \\
L_{71} &= \sqrt{k_{71}} |1\rangle\langle 7|, & L_{72} &= \sqrt{k_{72}} |2\rangle\langle 7|, & L_{73} &= \sqrt{k_{73}} |3\rangle\langle 7|,
\end{aligned} \tag{3.7}$$

where k_{ij} are the transition rates between states i and j . These include spin-preserving optical cycles (e.g., L_{14} , L_{41}) and non-radiative inter-system crossings (e.g., L_{47} , L_{71}).

Table 5 compares four parametrizations from Duarte *et al.*¹³, differing in decay rates (γ), spin flip probabilities (e), and ISC rates ($k_{ms,S}$, $k_{S,ms}$), in our simulations column 2 of the table was used, without the spin flip term (e). To maintain consistency between the inter-system crossing rates from Duarte *et al.*¹³ and the numbering, we chose that a re-indexing was needed as follows:

- $k_{ms,S} \rightarrow k_{n7}$: Transitions from excited triplet ($n = 4 (m_s = 0), 5 (m_s = -1), 6 (m_s = +1)$) to singlet
- $k_{S,ms} \rightarrow k_{7j}$: Transitions from singlet to ground triplet ($j = 1 (m_s = 0), 2 (m_s = -1), 3 (m_s = +1)$)

and $\gamma = k_{41} = k_{52} = k_{63}$ is the spontaneous decay rate. This systematic renumbering

Table 5 – Transition rates for spontaneous decay ($\gamma = k_{41} = k_{52} = k_{63}$), spin-flip parameter (e), and inter-system crossings (ISC) (renamed using the 1–7 level numbering) across four models from Duarte *et al.*¹³, *i.e.*, these are four possible set of values to use as transition rates found in this paper¹³. Since this paper is our model 2 and it compare 4 sets of values of transitions rates we choose to name each set as Model 2 – i , where for each i there is a set of values, in the original paper they called it Model i .

	Model 2 – 1	Model 2 – 2	Model 2 – 3	Model 2 – 4
γ (MHz)	77	62.7	63.2	67.4
e	1.5/77	0.01	0	0
k_{47} (MHz)	0	12.97	10.8	9.9
k_{67} (MHz)	30	80	60.7	91.6
k_{57} (MHz)	30	80	60.7	91.6
k_{71} (MHz)	3.3	3.45	0.8	4.83
k_{73} (MHz)	0	2.16/2	0.4	2.11/2
k_{72} (MHz)	0	2.16/2	0.4	2.11/2

Source: Adapted from DUARTE *et al.*¹³.

scheme, reflected in Fig. 3 and Table 5. In this table is important to notice that there are sets of transition rates from models 2 – 1 through 2 – 4, the values we choose to use were from model 2 – 2 of this table, where in the simulation code the option to switch to the other values was let available, the choice of writing model 2 – i is because we use this article as the basis of model 2 described ahead and there they have 4 different sets of transition rates which they name as model 1 to 4 here we use i to represent the same values found there. This preserves the original physical interpretation from the paper while aligning with our model's structure of optical processes.

The three terms in the first row of equation 3.7 (L_{14} , L_{25} and L_{36}) are related to the optical excitation, there $k_{14} = k_{25} = k_{36} = W_P$, where W_P is the optical pumping

rate, which is related to the laser power (P_L) via ²¹

$$W_P = c_L P_L, \quad (3.8)$$

where c_L depends on the photon emission statistics of each experimental setup, which can be determined by the saturation curve. A possible value found in Shandilya *et al.*²⁷ for $c_L = 10 \text{ MHz/mW}$, but W_P was set directly in the simulations.

The Hamiltonian model proposed at Magaletti *et al.*¹¹, which is included in the formulation of Eqs.3.6, presents an oversimplified description of the nitrogen-vacancy (NV) center, treating it as a two-level system. In this framework, the ground state corresponds to $m_s = 0$, while the excited state is represented by $m_s = -1$. The Hamiltonian for this reduced model is

$$H^{\text{M1}} = \frac{\Omega_R}{2} \left(|0\rangle \langle -1| e^{i\omega t} + |-1\rangle \langle 0| e^{-i\omega t} \right), \quad (3.9)$$

where Ω_R is the Rabi frequency and ω is the driving frequency.

Although this simplified approach provides a foundation for studying optical transitions, it exhibits critical limitations. First, the model does not reproduce spin dynamics under MW fields detuned from the resonant frequencies $\omega_0 = D_{\text{GS}} \pm \gamma_e B_0$. Second, it neglects hyperfine (HF) interactions and other multi-state couplings inherent to the triplet structure of the NV center. An *ad hoc* attempt to incorporate HF terms results in the modified Hamiltonian:

$$H_{\text{HF}}^{\text{M1}} = \frac{\Omega_R}{2} \left(|0\rangle \langle -1| e^{i\omega t} + |-1\rangle \langle 0| e^{-i\omega t} \right) \otimes \mathbb{I}_n + A_{\parallel} S_z I_z + A_{\perp} (S_x I_x + S_y I_y), \quad (3.10)$$

where $\mathbb{I}_n$ is the identity operator in the nuclear spin space and $\otimes$ denotes the tensor product.

However, this extension is too simplistic and can lead to nonsensical simulation results. The hyperfine interaction does not act solely in the two-level subspace $\{|0\rangle, |-1\rangle\}$, but couples all three spin states ($m_s = 0, \pm 1$) of the ground-state triplet of the NV center. Even with a simplified HF term $S_z I_z$, the restricted Hilbert space of model 1 artificially amplifies the effect of the HF interaction on optical dynamics. This inconsistency arises because the nuclear spin operator I_z acts across the entire triplet manifold, which is truncated in the two-level approximation. Consequently, predictions from this modified model deviate significantly from experimental observations and full theoretical treatments, such as Ernst *et al.*²⁵, Duarte *et al.*¹³ and Doherty *et al.*¹², seem more appropriate.

3.1.2 Model 2: Duarte *et al.*

This model uses the same base Hamiltonian shown in the beginning of the chapter, but the model for $H_{j,\text{HF}}$ is done as in Duarte *et al.*¹³, which describes a model where the hyperfine tensor's components depend on its spatial orientation through polar (θ_n) and azimuthal (φ_n) angles, depicted in Fig. 12. This model incorporates contact contributions

Table 6 – Hyperfine interaction components (MHz) for ^{15}N , carbon (C), and hydrogen-like (H) nuclear spins in the ground (GS) and excited (ES) states.

Family, State	$A_{\parallel}$	A_{ani}	$A_{\perp}$	$A'_{\perp}$
^{15}N , GS	3.4	0	7.8	0
^{15}N , ES	-58.1	0	-77	0
C , GS	-8.822	-0.789	-20.378	0.621
C , ES	-3.78	0.749	-14.12	0.680
H , GS	1.933	-0.250	2.067	0.670
H , ES	3.413	-0.349	4.086	0.866

Source: Adapted from DUARTE *et al* ¹³.

(A_c) and dipole-dipole hyperfine contributions (A_d), which decay exponentially and as $1/r^3$, respectively. The hyperfine Hamiltonian of this model is

$$\begin{aligned}
H_{j,\text{HF}}^{\text{M2}} = & A_{\parallel}^{(j)} S_z I_z + \frac{A_{\perp}^{(j)}}{4} (S_+ I_- + S_- I_+) \\
& + \frac{A'_{\perp}{}^{(j)}}{4} (S_+ I_+ e^{-2i\varphi_n} + S_- I_- e^{2i\varphi_n}) \\
& + \frac{A_{ani}^{(j)}}{2} [(S_+ I_z + S_z I_+) e^{-i\varphi_n} \\
& + (S_- I_z + S_z I_-) e^{i\varphi_n}],
\end{aligned} \tag{3.11}$$

where the coefficients $A_{\parallel}^{(j)}$, $A_{\perp}^{(j)}$, $A_{ani}^{(j)}$, and $A'_{\perp}{}^{(j)}$ depend on contact contributions (A_c), dipole-dipole hyperfine contributions (A_d) and θ_n , the polar angle of the nuclear spin relative to the NV quantization axis, and their dependence on these terms is stated in Eqs. 3.12. The coefficient values are listed in Table 6 for ^{15}N , carbon (C) and hydrogen-like environments (H), reflecting distinct coupling strengths in the ground and excited states.

$$A_{\parallel}^{(j)} = A_c - A_d(1 - 3\cos^2(\theta_n)) \tag{3.12a}$$

$$A_{\perp}^{(j)} = 2A_c + A_d(1 - 3\cos^2(\theta_n)) \tag{3.12b}$$

$$A'_{\perp}{}^{(j)} = 3A_d \cos(\theta_n) \sin(\theta_n) \tag{3.12c}$$

$$A_{ani}^{(j)} = 3A_d \sin^2(\theta_n) \tag{3.12d}$$

3.1.3 Model 3: Doherty *et al*.

This model still has the same Hamiltonian as in the beginning of the chapter and is presented in the work of Doherty *et al*.¹², where the hyperfine interaction depends only on the coefficients $A_{\parallel}^{(j)}$, $A_{\perp}^{(j)}$, which depend on A_c and A_d , but in a simpler way than Model 2, where $A_{\parallel}^{(j)} = A_c + 2A_d$ and $A_{\perp}^{(j)} = A_c - A_d$. The interaction term is given by

$$H_{j,\text{HF}}^{\text{M3}} = A_{\parallel}^{(j)} S_z I_z + A_{\perp}^{(j)} (S_x I_x + S_y I_y), \tag{3.13}$$

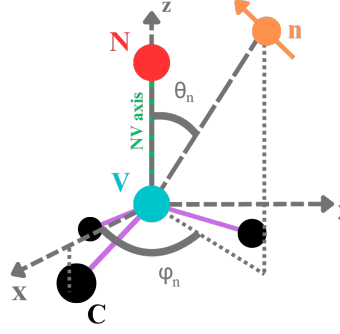


Figure 12 – Cartesian representation of the NV center with a nearby nuclear spin. Here, the angles θ_n and φ_n are the polar and azimuthal angles of the nuclear-spin position vector relative to the NV quantization axis.

Source: Adapted from DUARTE *et al* ¹³.

this hyperfine term is usually what appears in most works ^{5, 54, 10, 51, 67}. The constants are shown in Table 4. It is important to remember that all these Hamiltonians are for the room temperature energy level structure of NVs.

3.2 Our present model in QuTiP

Before presenting our model, it is useful to have a brief overview of the Library/Package we will be using and its capabilities.

3.2.1 What is QuTiP

The Quantum Toolbox in Python(QuTiP) is an open source library in Python that, throughout its history, has been developed to be a multipurpose tool for studying quantum systems. It has functionalities ranging from quantum control and information processing to simulation of quantum circuits and open quantum systems. It was first envisioned to solve problems in quantum optics simulations, but, being open-source developed, it was adopted and maintained by the scientific community and turned into a versatile tool. It has complementary library modules that are used for control (QuTiP-QOC) and circuit simulation (QuTiP-QIP), and also for parallelized GPU process of simulations (QuTiP-JAX). Furthermore, the same organization has written a version of QuTiP in Julia, which is in early stages of development, `QuantumToolbox.jl`. QuTiP is used in many projects, open source or not (e.g. Qiskit, Nvidia CUDA-Q), as a visualization or even backend tool. Its versatility comes not only from the range of possible applications it has, but also from the fact that it can be used to instantiate custom problems, using its structure to tackle very complex systems by using a class of solvers for some different ways of defining quantum systems, such as ^{4, 3, 35}

- Lindblad master equation
- Monte-Carlo master equation
- Bloch-Redfield master equation
- Floquet master equation
- Krylov subspace solver
- Stochastic master equation
- Hierarchical equations of motion
- Permutational invariant systems

These are all solvers defined in the Solver Class of `Qutip`, and more than that, it is possible to define other solvers on top of it and all of them can be used, either with the optimization methods defined in `QuTiP-QOC` for defining optimal control pulses, or with other control techniques largely available or developed by each user. `QuTiP` can easily instantiate states, Hamiltonians (static or not), and Lindblad operators, or even define stochastic systems or baths that will interact with the target systems. It has methods defined in the proper class for calculating quantities very important for analyzing solutions to various quantum problems, such as fidelity, emission spectrum, and partial trace, just to cite a few. One extra point is that, although using superoperator representation as default, it can switch between Choi, Kraus, Stinespring, and χ representations.^{4, 3, 35}

This versatility allows anyone to create their library on top of it, and with that taken into account, choosing this package seemed like a natural choice. Since it makes possible to expand a single specific problem into any or all directions wanted, given that one of the goals is to be able to characterize (or automatize characterization) and do precise/optimal control, employing machine learning techniques, with the system in lab; being able to assemble a simulation environment where, adding and removing parts as need, and adjusting parameters easily are important aspects for reaching these goals.

3.2.2 What is new in our model

The model we used is a combination of the optical transition present in Model 1 and some other models, but since this model does not care about how the transitions due to MW field happen and how the magnetic field affect those transitions we combined it with the transitions that happen due to the static Hamiltonian present in both Model 2 and 3, where Model 2 allows for different types of hyperfine interactions other than Nitrogen to be studied. We also use a simplified version of models 2 and 3 without hyperfine interaction, which turns them into the same model, so that we can see the effects of this interaction on the system.

While our explicit modeling of the hyperfine interaction captures one coherent environmental coupling, this treatment alone cannot account for the full dissipative dynamics observed experimentally. A complete description requires us to address: (1) phonon-induced orbital relaxation, (2) charge fluctuations in the diamond lattice, (3) multi-spin correlations with the spin bath of the possible spins (P1 centers, C, other NVs, N, etc), and (4) paramagnetic defect interactions⁵, all aspects that are possible to be implemented using other solvers or defining a customized solver in QuTiP. Here we will incorporate the dissipative effects via Lindblad operators, which are used to take into account both relaxation (T_1) and dephasing (T_2^*) mechanisms in the ground and excited states, which are expressed as^{21, 18}

$$\begin{aligned} L_{rel,-,GS} &= \sqrt{\frac{1}{2T_{1,GS}}} S_{-,GS}, & L_{rel,+,GS} &= \sqrt{\frac{1}{2T_{1,GS}}} S_{+,GS}, & L_{dep,GS} &= \sqrt{\frac{1}{T_{2,GS}^*}} S_{z,GS}, \\ L_{rel,-,ES} &= \sqrt{\frac{1}{2T_{1,ES}}} S_{-,ES}, & L_{rel,+,ES} &= \sqrt{\frac{1}{2T_{1,ES}}} S_{+,ES}, & L_{dep,ES} &= \sqrt{\frac{1}{T_{2,ES}^*}} S_{z,ES}, \end{aligned} \quad (3.14)$$

where $T_{1,GS}$ and $T_{1,ES}$ denote the longitudinal spin relaxation times, while $T_{2,GS}^*$ and $T_{2,ES}^*$ represent the pure dephasing times for the ground and excited states, respectively. Typical parameter values for NV centers under ambient conditions are $T_{1,GS(ES)} \approx 1$ ms, $T_{2,GS}^* \approx 3$ μ s, and $T_{2,ES}^* \approx 6$ ns following Duarte *et al.*¹³, not to be confused with the optical lifetime of ES (≈ 10 ns). That being said, with these tools at hand, we can analyze a first implementation of these models in QuTiP and see how well we can reproduce other literature results in either experimental or simulation fashion.

4 RESULTS AND DISCUSSION

In this chapter, we present the results regarding the simulations. We show that it is possible to reproduce with some accuracy the behaviors of experimental setups and other simulations. We also want to know what the limitations of our simulations are, and what parameters we need to be aware of for reproducing well the desired behaviors. In section 4.1, we see how the population of the 7 levels presented earlier behaves. These population behaviors can be seen only in simulated environments, so for that we compare them with other simulations to have an idea of how accurate our simulations are, and we use them to see which of the models behaves more closely to what is expected. In section 4.2, we see how the PL behaves and how magnetic field strength and orientation, and also MW strength, affect the PL. In section 4.3, we see how well we can reproduce the outcomes of the different protocols described in Sections 2.3 and 2.4, using the models presented in Chapter 3. We finish by presenting, in section 4.4, the results regarding the performance of our simulations, using as a benchmark the execution time of a simple protocol run using models 2, 3 and the models without HF interaction.

4.1 Results: Population Dynamics

Here, we present the results related to the population dynamics of the simulations of the NV center using the QuTiP package, according to a test sequence (named sequence 1) adapted from Magaletti *et al.*¹¹, which consists of three steps: the first one is a population evolution under the influence of a laser for $10\ \mu\text{s}$, followed by a free evolution so that the population can transfer from the ISC $|7\rangle$ state to the ground states, finishing with a evolution under the influence of a MW field for $1\ \mu\text{s}$ with strength Ω_R (Rabi frequency) and driving frequency ω . In Fig. 13b, we are using model 1 and sequence 1, repeated 5 times, to reproduce the results of Magaletti *et al.*¹¹, which are presented for easy comparison in Fig. 13a, from the original paper.

In Figure 13, the authors of Magaletti *et al.*¹¹ used $\Omega_R = 15\ \text{MHz}$, we have used $\Omega_R = 15.7\ \text{MHz}$. In their paper, they do not provide a value for ω explicitly, so we empirically searched for a value that could reproduce their results, and what we used was $\omega = 0.1\ \text{MHz}$. When this value is set to zero, the values of n_c , described in the original paper as the coherence between the oscillating states (here, n_1 and n_2), stay flat at zero during all steps of evolution. If we increase it above one, the populations between states that oscillate during MW influence do not get fully transferred, and values between zero and one change a little bit the phase in which the populations oscillate. The value that better reproduced the original paper was the one we chose. An interesting behavior to note is that after optical excitation and free decay time to prepare the state n_1 , we achieve

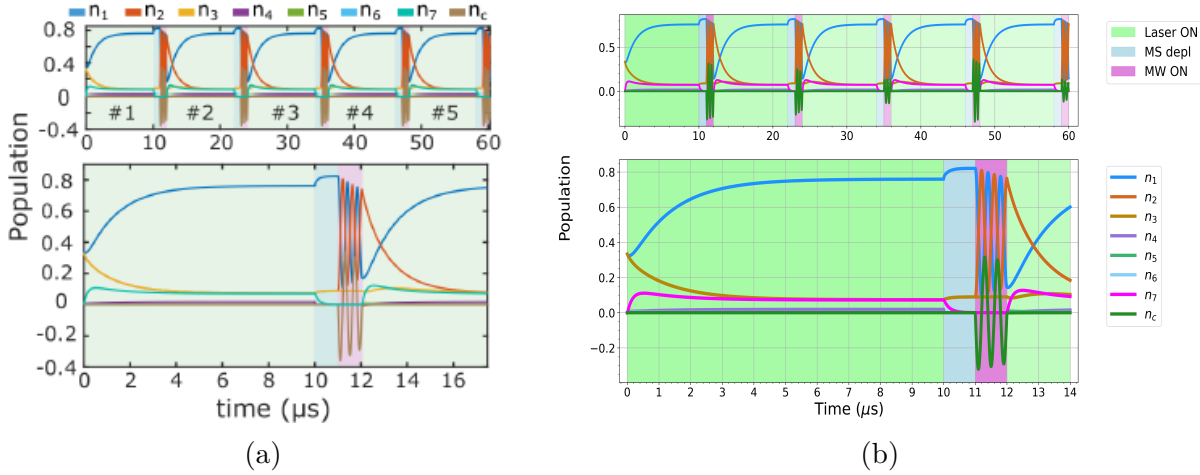


Figure 13 – Comparison between the results reported by Magaletti *et al.*¹¹ and our QuTiP simulations for the same protocol, showing good agreement: (a) original figure from Magaletti *et al.*¹¹; (b) is our simulation using QuTiP, where the green regions indicate laser power ON, the blue regions are free evolution, and the magenta regions are for MW power ON.

Source: (a) Adapted from MAGALETTI *et al.*¹¹.

Source: (b) By the author.

a total population of approximately 0.8, *i.e.*, the state gets 80% of the total population available. The rest of the population is equally divided between the states n_2 and n_3 . Another observation is that the oscillations n_c in the original paper are similar in all five repetitions, but change in each repetition after the first in our simulations.

4.1.1 Hyperfine effect

Here, we investigate the effects of the hyperfine interaction. To do that, we altered the sequence 1 used, changing the time evolution under the influence of the MW field to $4.5 \mu\text{s}$, and we also removed the repetitions. These two changes were made to have more detailed visuals of what is happening during the full evolution, and the extended time was used to have more details in the MW excitation. First, we look at the naive addition of the hyperfine interaction term of model 3 to model 1, it is good to point out and remember that model 1 never deals with hyperfine interactions or even magnetic field interactions; it assumes that these elements are well handled and would not play a role during the protocol.

It is possible to see in Fig. 14 that adding naively the hyperfine term used in model 3, *i.e.*, the one from Eq. 3.13, produces a huge suppression of population transfer during optical excitation and during MW driving. To verify if this result was what is supposed to happen or it just happened because of the definition of the model due to its assumptions, we turned to the other two models and see how they behave when we use their HF terms

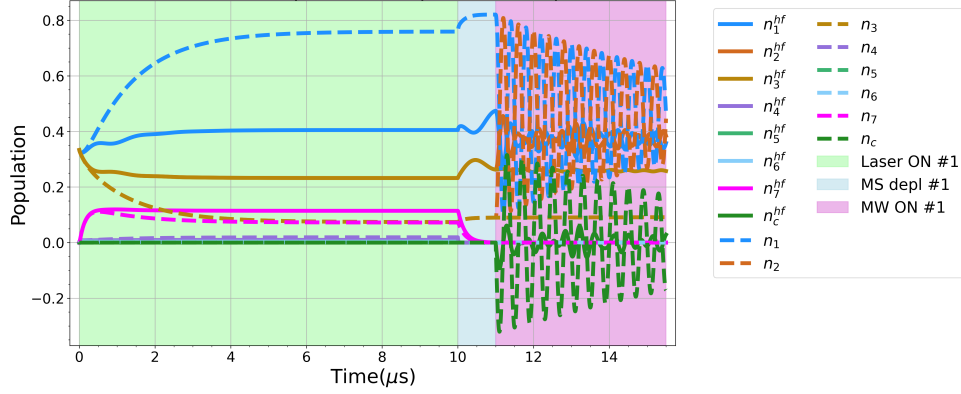


Figure 14 – Population evolution using the sequence 1 (with $\tau_{\text{MW}} = 4.5 \mu\text{s}$) with naive added hyperfine terms to the model 1 (—) compared to the same model without HF terms (---). The color scheme of the regions is the same as Fig. 13b

Source: By the author.

in the case with no HF terms (models 2 and 3 have the same Hamiltonian when using no HF terms). Fig. 15 shows the difference between hyperfine and non-hyperfine interactions being taken into account. The HF terms used for these plots are those from Eq. 3.13 (model 3), and the static magnetic field is 100 G.

Figure 15 shows that the hyperfine interaction influences the evolution of MW when Ω_R is below a threshold (which we have not determined but seems to be close to 4 MHz), but when it is high enough, there is no visible difference in the dynamics. A behavior that does not match what happened when using model 1 with a naive addition of HF terms.

Using the same sequence, we plotted the strength of the static magnetic field ($B_0 = \|\vec{B}_0\|$) and the MW field strength (Ω_R). The comparison between HF without HF and model 3 is shown in Figs. 16 and 17.

Figure 16 shows that when $B_0 = 0$ G, the evolution under MW driving of states n_2 and n_3 always has the same population, making the population of state n_1 decrease faster than when there is a static magnetic field present (Figs. 15 and 17). The mean and standard deviation values of the population difference between model 3 and no HF are presented in Table 7 as well as the percentage change(Δ) of the population differences in each evolution step between $\Omega_R = 20.94$ MHz and 2.09 MHz.

We can also see that there is a small difference in population transfer during laser excitation when B_0 is close to 500 G there is a small increase in the state population n_2 with a decrease in n_1 compared to the population without HF, this behavior can be seen in Fig. 17 and the value of the mean and standard deviation difference of the population between the non-HF and model 3 populations of each level in each step of the evolution can be seen in Table 8.

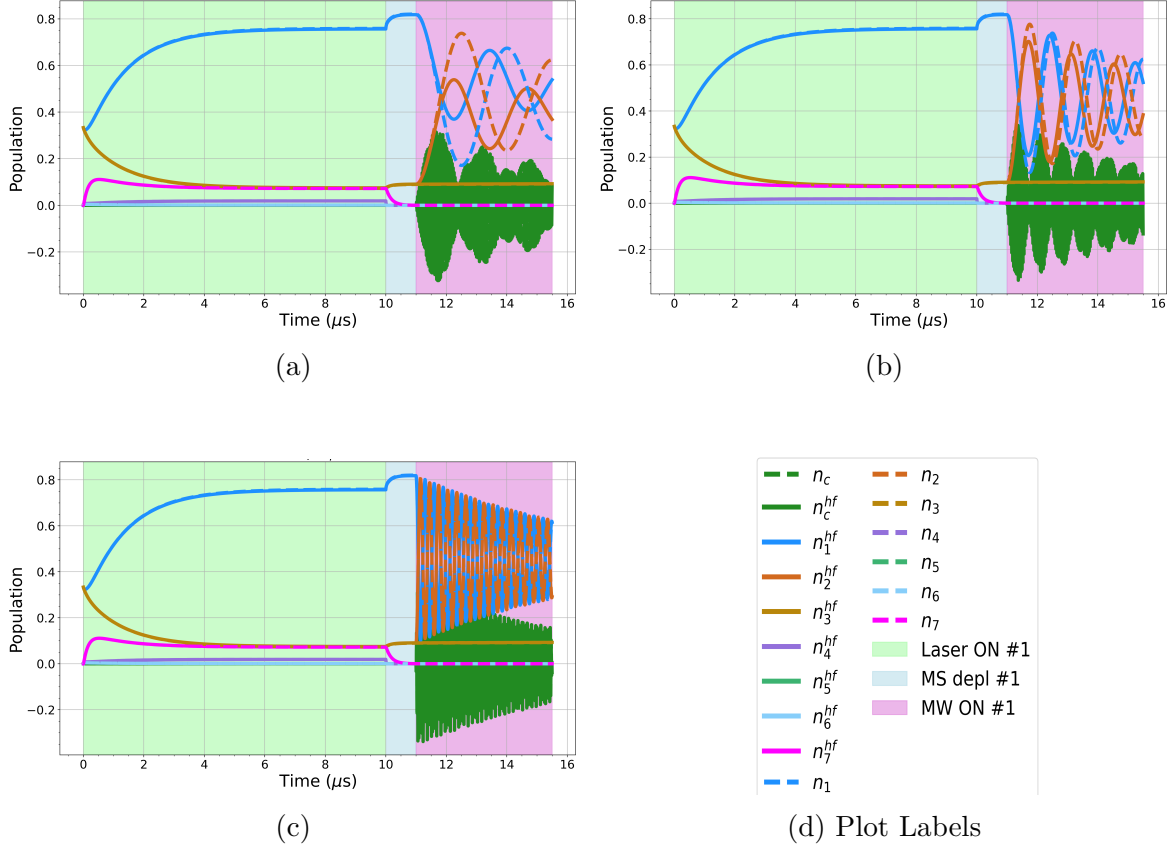


Figure 15 – Comparison between population dynamics in the hyperfine (—) and non-hyperfine (--) interaction of the model presented in Doherty *et al.*¹², the static magnetic field is set to $B_{0,z} = 100$ G: (a) $\Omega_R = \pi/T_{2,GS}^* \approx 2.09$ MHz, (b) $\Omega_R = 2\pi/T_{2,GS}^* \approx 4.19$ MHz, (c) $\Omega_R = 10\pi/T_{2,GS}^* \approx 20.94$ MHz. Sequence 1 was used here with the modification of setting $\tau_{MW} = 4.5 \mu s$, in all we kept $W_P = 1.9$ MHz.

Source: By the author.

We used the same sequence and values of B_0 and Ω_R from the previous plots to see if there were differences in the population between hyperfine models 2 and 3 models. The differences exist but are very small and observed only during the evolution of the MW for low Ω_R , as depicted in Fig. 18 with the values of the mean and standard population difference between models 2 and 3 presented in Table 9.

4.2 Results: Photoluminescence (PL)

Here, we present the results for the behavior of the PL. For that, we use state preparation sequences for seeing how the PL changes when we prepare n_1 , n_2 and $(n_1 + n_2)/2$, these sequences are done by applying the laser for $10 \mu s$ and doing a free evolution for $1 \mu s$, so the same as the first two parts of sequence 1, after this there are three paths: do nothing, for preparing n_1 , apply a π -pulse for preparing n_2 or a $\pi/2$ -pulse for preparing the superposition state; the sequence finishes with a laser being applied for $\tau_F 10 \mu s$, the

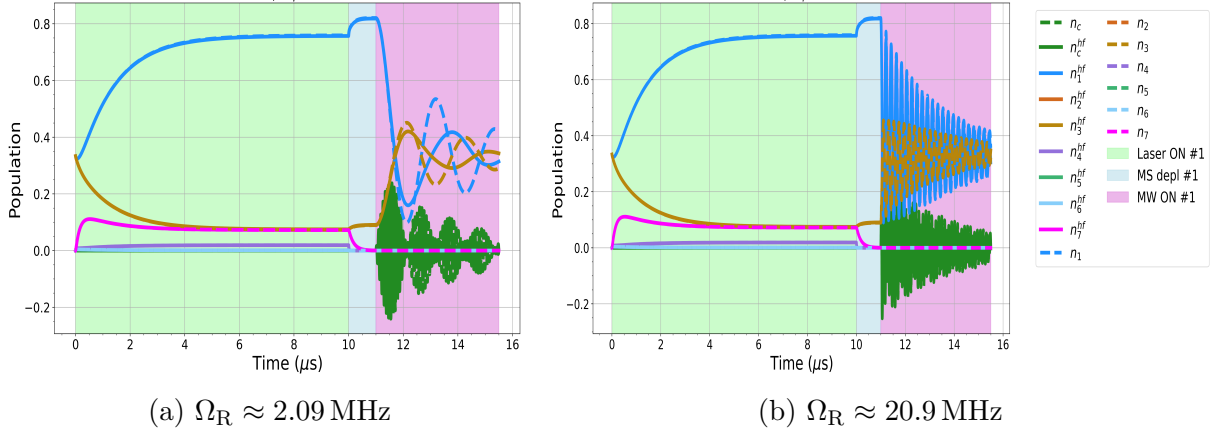


Figure 16 – Comparison between population dynamics in hyperfine (—) and no hyperfine interaction (--) of the model 3, the static magnetic field is set to $B_{0,z} = 0$ G and (a) $\Omega_R = \pi/T_{2,GS}^* \approx 2.09$ MHz, (b) $\Omega_R = 10\pi/T_{2,GS}^* \approx 20.94$ MHz. Here sequence 1 had $\tau_{MW} = 4.5 \mu s$, in all we kept $W_P = 1.9$ MHz. Low and high values for Ω_R have an impact on how much the behavior of the hyperfine interaction influences the dynamics, here at no magnetic field the states $m_S = \pm 1$ keep always the same population in between them.

Source: By the author.

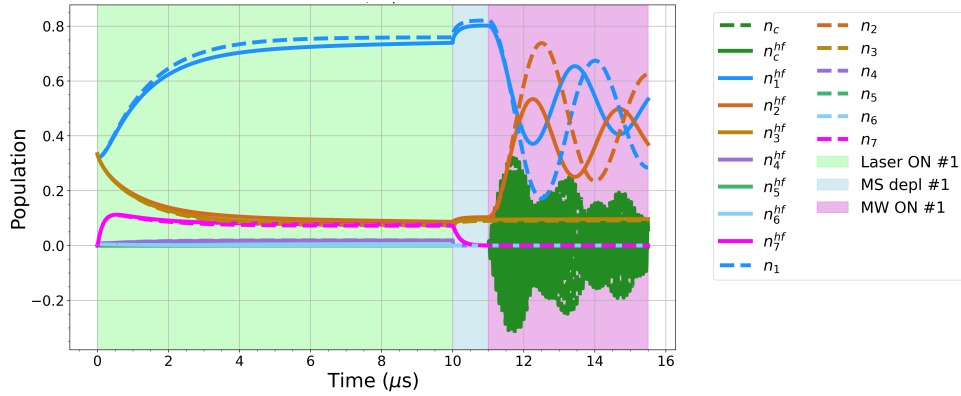


Figure 17 – Plot of the comparison between model 3 with (—) and without (--) hyperfine interaction for $B_0 = 500$ G and $\Omega_R \approx 2.09$ MHz. Showing the difference in the MW regime and a small difference in the laser regime for this value of magnetic field due to ESLAC.

Source: By the author.

readout time can be this full time or a part of it in the beginning where it is integrated, τ_F is a time enough for preparing a new initialization protocol. We then look at the PL intensity of the last step and plot it over the time duration t_F . We also vary the strength of the static magnetic field and the MW field strength. In all cases, we are using the driving frequency ω as the resonant frequency of the NV center with state $m_S = -1(n_2)$, *i.e.*, $\omega = D_{GS} - \gamma_e B_{0,z}$.

Table 7 – Comparison of $\Delta n_i = |n_i^{(M3)} - n_i^{(\text{NoHF})}|$ values for Laser, Free, and MW protocols at $\Omega_R = 20.94 \text{ MHz}$ and 2.09 MHz , here $B_0 = 0 \text{ G}$. Laser and Free values are unchanged, so only one column is shown. $\% \Delta$ is the percent change of Δn_i between different Ω_R . All values are with a 10^{-3} factor multiplied, except for $\%$ values, we didn't use scientific notation so that the values could be compared in terms of scale between evolution stages

n_i	Laser		Free		MW		
	Value	$\% \Delta$	Value	$\% \Delta$	20.94	2.09	$\% \Delta$
<i>Mean Absolute Difference</i>							
1	1.643	0.000	2.050	0.000	30.965	82.834	167.508
2	0.684	0.000	0.989	0.000	15.483	41.417	167.500
3	0.684	0.000	0.989	0.000	15.483	41.417	167.500
4	0.070	0.000	0.002	0.000	0.000	0.000	–
5	0.017	0.000	0.000	–	0.000	0.000	–
6	0.017	0.000	0.000	–	0.000	0.000	–
7	0.311	0.000	0.073	0.000	0.000	0.000	–
<i>c</i>	0.000	–	0.000	–	21.722	53.095	144.420
<i>Standard Deviation</i>							
1	0.642	0.000	0.058	0.000	35.856	101.954	184.343
2	0.274	0.000	0.019	0.000	17.928	50.977	184.343
3	0.274	0.000	0.019	0.000	17.928	50.977	184.343
4	0.022	0.000	0.009	0.000	0.000	0.000	–
5	0.005	0.000	0.002	0.000	0.000	0.000	–
6	0.005	0.000	0.002	0.000	0.000	0.000	–
7	0.107	0.000	0.099	0.000	0.000	0.000	–
<i>c</i>	0.000	–	0.000	–	25.255	65.239	158.333

Source: By the author.

In Fig. 19 is presented the profiles of PL for readout of states $|0\rangle$ (blue/top line), $(|0\rangle + |1\rangle)/\sqrt{2}$ (orange/middle line) and $|1\rangle$ (dark gold/bottom line), in two different regimes, when HF interaction is not present (Fig. 19a) and with it present using model 3 (Fig. 19b), also it is compared the readout when the states are prepared through the usual procedure stated before (—) and setting the states as pure states (--) and just doing the last part of the procedure. The mean absolute difference in PL between each prepared and pure state is in Table 10.

In Fig. 20 is presented the profiles of PL for readout of states $|0\rangle$ (blue/top line) and $|1\rangle$ (orange/middle line), and the contrast between the PL of these states (green/bottom line), in two different regimes, when HF interaction is not present (Fig. 19a) and with it present (Fig. 19b), the maximum $\%$ contrast of them are 38.12% and 37.55%, respectively, the last value goes near 34.56% for HF when near ESLAC, the plots in Figs. 19 and 20 are using $B_0 = 100G$ just for an example of PL and Contrast profile, here we normalized PL for the maximum value of state $|0\rangle$ in the time interval.

Table 8 – Absolute differences ($\Delta n_i = |n_i^{(M3)} - n_i^{(\text{NoHF})}|$), with $B_0 = 500$ G and $\Omega_R \approx 2.09$ MHz, for each protocol region and error statistics. All values are with a 10^{-3} factor multiplied, except for % values, we didn't use scientific notation so that the values could be compared in terms of scale between evolution stages

n_i	Laser		Free		MW	
	Mean Abs Diff	Std Dev Diff	Mean Abs Diff	Std Dev Diff	Mean Abs Diff	Std Dev Diff
1	23.676	7.770	18.938	0.516	126.044	141.091
2	15.481	5.075	13.528	0.180	127.240	141.080
3	4.297	1.678	4.761	0.167	5.032	0.116
4	0.993	0.260	0.016	0.081	0.000	0.000
5	0.414	0.104	0.005	0.028	0.000	0.000
6	0.061	0.023	0.001	0.004	0.000	0.000
7	4.416	1.264	0.659	0.886	0.001	0.002
c	0.000	0.000	0.000	0.000	150.925	166.470

Source: By the author.

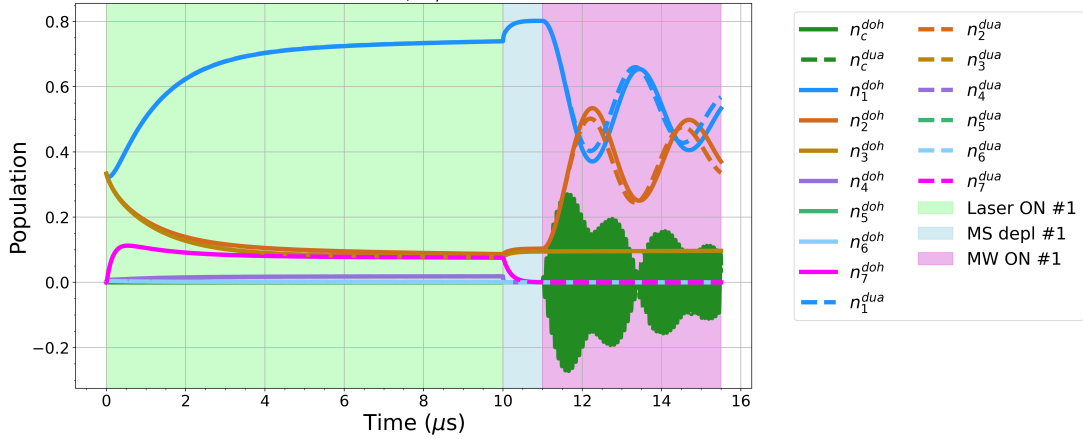


Figure 18 – Plot of the comparison between model 2 and model 3 with $B_0 = 500$ G and $\Omega_R \approx 2.09$ MHz. Showing the small difference in the MW regime and same behavior in the other regimes highlighting that the difference in the way HF is defined influences only the MW interactions.

Source: By the author.

In Fig. 21 is presented how the PL varies in the readout when changing the optical pumping rate (W_P) while preparing (using optical pumping) states $|0\rangle$ (Fig. 21a) and $|1\rangle$ (Fig. 21b). The bottom line (blue) is for $W_P = 0.5$ and the top line (gray) is for $W_P = 4$, and the lines between are for a variation of 0.5 in W_P , each step is the next closest line; here we reproduce what is seen in Fig. 5.

In Fig. 22, we compare the PL when preparing the states $|0\rangle$ (blue/top line, $(|0\rangle + |1\rangle)/\sqrt{2}$ (orange/middle line) and $|1\rangle$ (dark gold/bottom line) when using model 3 (Fig. 22a) and model 2 (Fig. 22b), both with a $B_0 = 0$ G and $\Omega_R \approx 2.09$ MHz. There are not many differences in the profile depending on the model chosen.

Table 9 – Absolute differences ($\Delta n_i = |n_i^{(M3)} - n_i^{(M2)}|$), with $B_0 = 500$ G and $\Omega_R \approx 2.09$ MHz, for each protocol region and error statistics. All values are with a 10^{-3} factor multiplied, except for % values, we didn't use scientific notation so that the values could be compared in terms of scale between evolution stages

n_i	Laser		Free		MW	
	Mean Abs Diff	Std Dev Diff	Mean Abs Diff	Std Dev Diff	Mean Abs Diff	Std Dev Diff
1	0.428	0.207	0.095	0.002	23.943	23.782
2	0.276	0.141	0.054	0.001	23.934	23.786
3	0.083	0.037	0.039	0.001	0.019	0.022
4	0.017	0.009	0.000	0.000	0.000	0.000
5	0.007	0.004	0.000	0.000	0.000	0.000
6	0.001	0.001	0.000	0.000	0.000	0.000
7	0.078	0.039	0.003	0.003	0.000	0.000
c	0.000	0.000	0.000	0.000	25.203	28.523

Source: By the author.

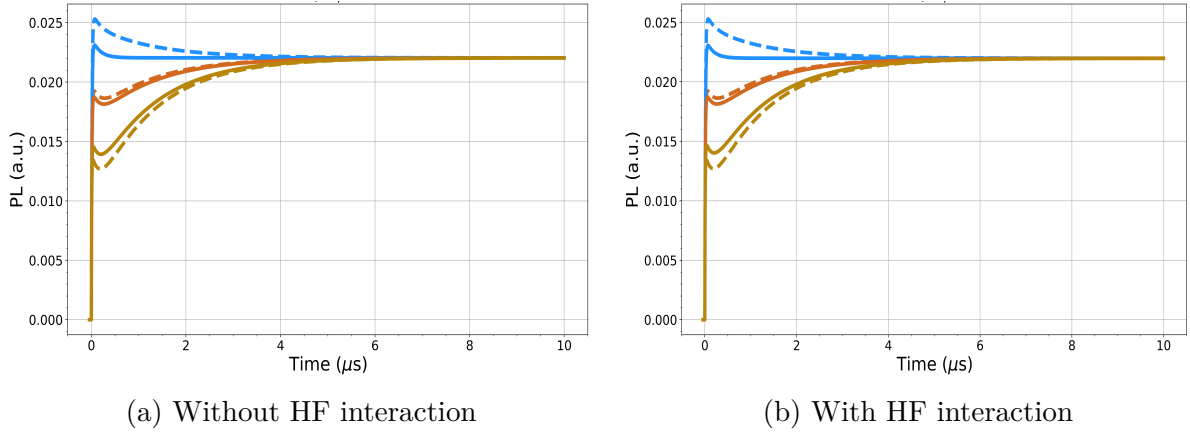


Figure 19 – PL of pure (--) versus prepared (—) state readout, where blue (top lines) is $|0\rangle$, orange (middle lines) is $(|0\rangle + |1\rangle)/\sqrt{2}$ and yellow (bottom lines) is $|1\rangle$

Source: By the author.

In Fig. 23, we compare the PL when preparing the states $|0\rangle$ (blue/top line), $(|0\rangle + |1\rangle)/\sqrt{2}$ (orange/middle (23a)/bottom (23b) line) and $|1\rangle$ (dark gold/bottom (23a)/middle (23b) line) when using model $\Omega_R \approx 2.09$ MHz (Fig. 23a) and $\Omega_R \approx 20.9$ MHz (Fig. 23b), both with a $B_0 = 0$ G and using model 3. We see an inversion of the signal between the prepared $|1\rangle$ state and the superposition state when Ω_R is below versus above the threshold mentioned in Sec. 4.1, in the absence of a magnetic field.

In Fig. 24, we compare the PL when preparing the states $|0\rangle$ (blue/top line), $(|0\rangle + |1\rangle)/\sqrt{2}$ (orange/middle (24a)/bottom (24b) line) and $|1\rangle$ (dark gold/bottom (24a)/middle (24b) line) when using model $B_0 = 0$ G (Fig. 24a) and $B_0 = 500$ G (Fig. 24b), both with $\Omega_R \approx 20.9$ MHz and using model 3. We see an increase in separation of the signal of $|1\rangle$ being prepared and the superposition state with the introduction of the magnetic field; the signal also does not pass through that inversion of the last figure when there is a

Table 10 – Comparison of absolute differences of PL from pure against prepared state in Model 3 with and without hyperfine interaction at $B = 100$ G, $\Omega_R = 15.70$ MHz.

Initial State	Mean Abs Diff		Std Dev Diff		Max Diff	
	HF	No-HF	HF	No-HF	HF	No-HF
$ 0\rangle$	$5.35 \cdot 10^{-4}$	$9.46 \cdot 10^{-4}$	$6.35 \cdot 10^{-4}$	$6.83 \cdot 10^{-4}$	$1.703 \cdot 10^{-3}$	$2.567 \cdot 10^{-3}$
$\frac{ 0\rangle + 1\rangle}{\sqrt{2}}$	$4.72 \cdot 10^{-4}$	$2.90 \cdot 10^{-4}$	$3.09 \cdot 10^{-4}$	$3.5 \cdot 10^{-5}$	$6.47 \cdot 10^{-4}$	$4.01 \cdot 10^{-4}$
$ 1\rangle$	$7.87 \cdot 10^{-4}$	$5.20 \cdot 10^{-4}$	$7.9 \cdot 10^{-5}$	$6.45 \cdot 10^{-4}$	$1.054 \cdot 10^{-3}$	$1.844 \cdot 10^{-3}$

Source: By the author.

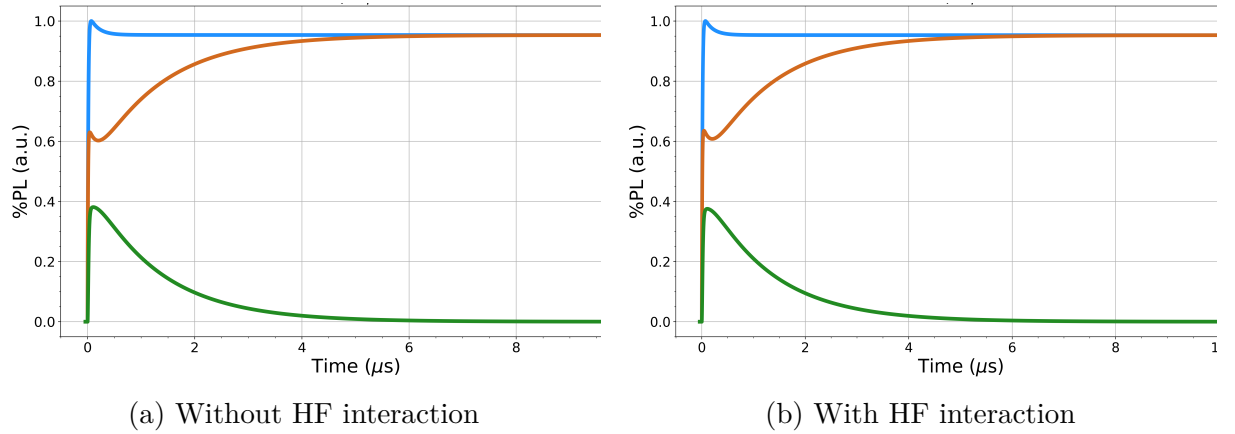


Figure 20 – Percentage PL of prepared state readout, where blue (top line) is $|0\rangle$, orange (middle line) is $|1\rangle$ and green (bottom line) is the contrast of these PLs

Source: By the author.

magnetic field present and Ω_R below or above the threshold mentioned before.

In Fig. 25, we explore the property in section 2.3 and shown in Fig. 7. This is the steady state PL under optical excitation against B_0 for varying the polar angle θ_B (each color represents an angle) of $\vec{B}_0$ in relation to the quantization axis of NV. The solid lines that are transparent were obtained from using Model 3 without the HF term, and the dashed lines with markers from using Model 3, we observe that the effect is the same and shows a drop in PL intensity of the steady state under laser illumination in the ESLAC and GSLAC regions and when the angle changes the drop in PL intensifies. However, the absolute magnitude of PL is different from that shown in Fig. 7. Nevertheless, this difference is, at least, partially expected because our curve is normalized to the simulated steady-state PL, using the parameters in Chapter 3, while Fig. 7 reports absolute experimental PL for a different sample and excitation conditions in Ref.¹⁸.

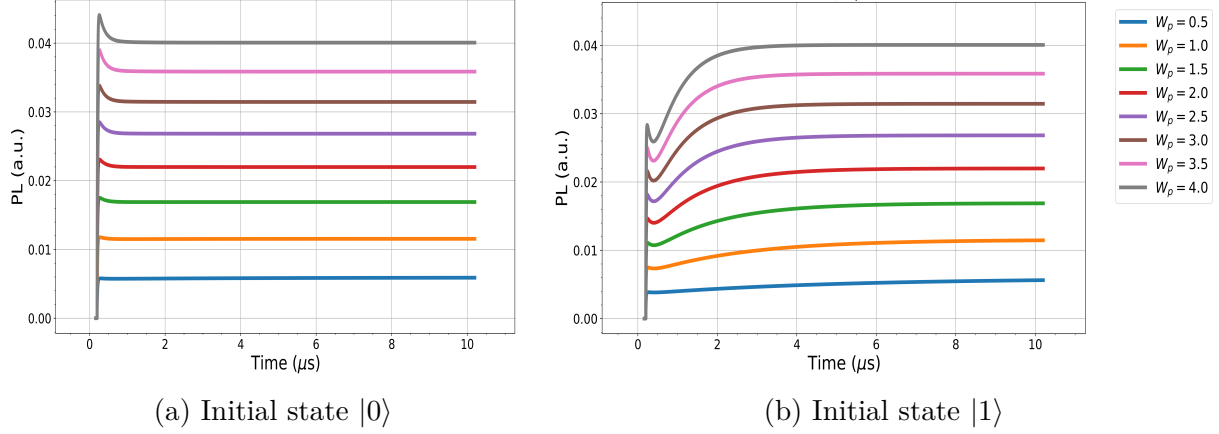


Figure 21 – Depiction of how varying the optical pumping rate (W_P) affects the PL intensity going from 0.5 (blue/bottom line) to 4 (gray/top line) in 0.5 steps. The values for static magnetic field intensity ($\|\vec{B}_0\|$) and Rabi frequency (Ω_R) where 100 G and 15.7 MHz respectively. Here we used model 3 HF term

Source: By the author.

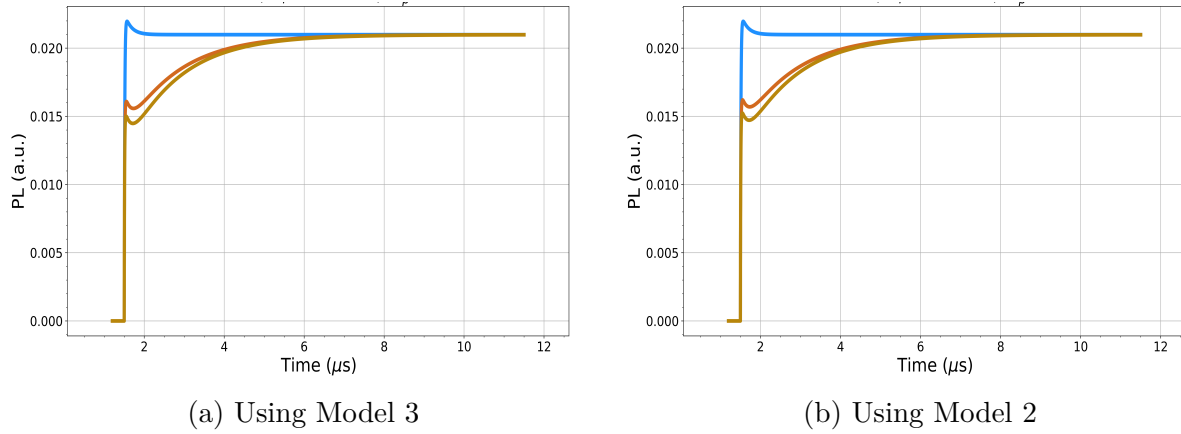


Figure 22 – Plots of the PL intensity of readout for states $|0\rangle$ (blue/top line), $(|0\rangle + |1\rangle)/\sqrt{2}$ (orange/middle line) and $|1\rangle$ (dark gold/bottom line) prepared via optical pumping in models 3 (a) and 2 (b) in $B_0 = 0$ G and $\Omega_R \approx 2.09$ MHz

Source: By the author.

4.3 Results: ODMR

In CW-ODMR and Pul-ODMR, we compare the results of models 2 and 3 with and without HF interactions, varying B_0 to see the effects of this parameter on PL, asserting that it follows experimental observation. In CW-ODMR and Pul-ODMR we have chosen that the readout has $W_P = 1.9$ and for the MW interaction part $\Omega_R \approx 15.7$ MHz, since values near 4 MHz are close enough to the threshold to promote the full transition between states.

We used two parametric functions to fit the simulated spectral features: a Lorentzian

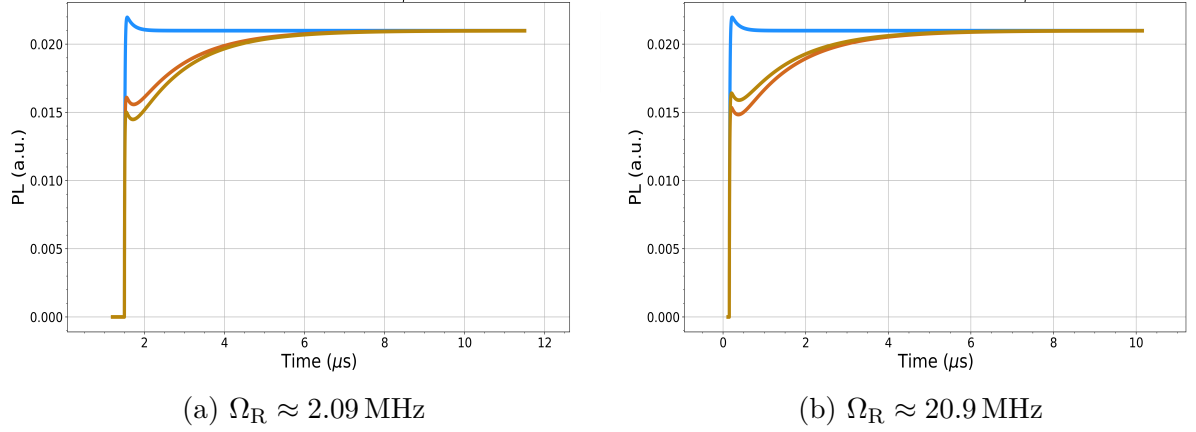


Figure 23 – Plots of the PL intensity of readout in states $|0\rangle$ (blue/top line), $(|0\rangle + |1\rangle)/\sqrt{2}$ (orange/middle (a)/bottom (b) line) and $|1\rangle$ (dark gold/bottom (a)/middle (b) line) prepared via optical pumping in model 3 for $B_0 = 0$ G, $\Omega_R \approx 2.09$ MHz (a) and $\Omega_R \approx 20.9$ MHz (b).

Source: By the author.

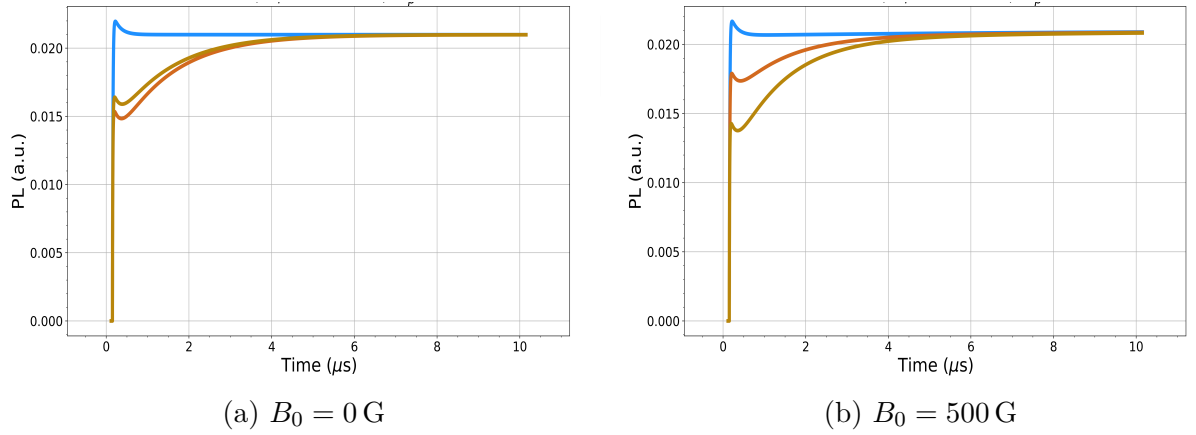


Figure 24 – Plots of the PL intensity of readout using states $|0\rangle$ (blue/top line), $(|0\rangle + |1\rangle)/\sqrt{2}$ (orange/middle (a)/bottom (b) line) and $|1\rangle$ (dark gold/bottom (a)/middle (b) line) prepared via optical pumping in model 3 for $\Omega_R \approx 20.9$ MHz, $B_0 = 0$ G (a) and $B_0 = 500$ G (b).

Source: By the author.

function and a generalized Gaussian function, both modified to represent inverted dips. The expressions for each are as follows.

$$\text{lorent}(x) = -A \cdot \frac{1}{\pi\gamma} \cdot \frac{1}{1 + \left(\frac{x-x_0}{\gamma}\right)^2} + D \quad (4.1)$$

$$\text{gen_gauss}(x) = -A \cdot \frac{\beta}{2\gamma\sqrt{\pi}\Gamma\left(\frac{1}{\beta}\right)} \cdot \exp\left(-\left|\frac{x-x_0}{\gamma}\right|^\beta\right) + D \quad (4.2)$$

In both expressions:

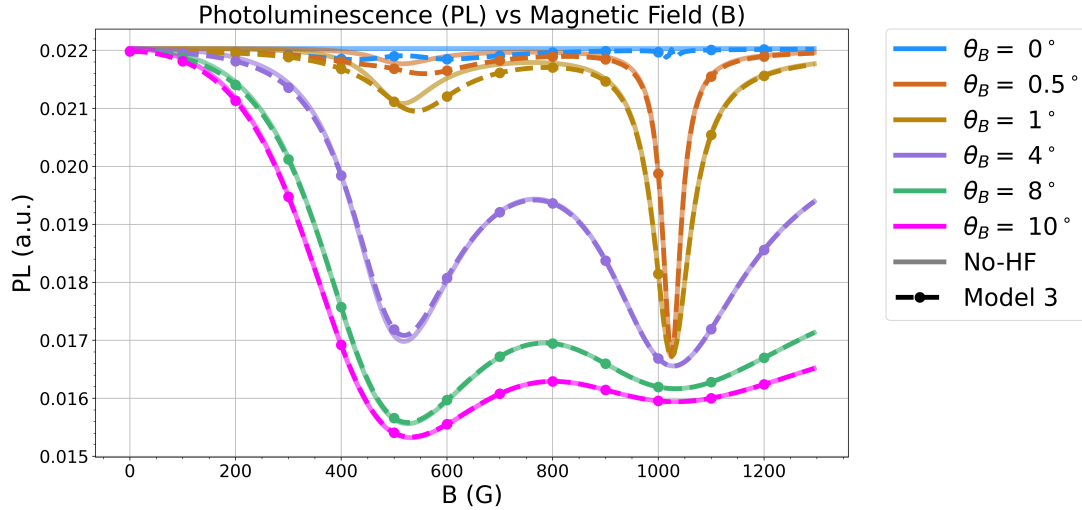


Figure 25 – PL intensity of the steady state of laser illumination while varying the static magnetic field intensity and the polar angle from the quantization axis of NV reproducing, to some extent, the effect from Fig. 7. Each color represents an angle choice for varying the B_0 , the dashed lines with dots on it are using Model 3 and the solid lines with some transparency is using Model 3 without HF interaction.

Source: By the author.

Table 11 – Fit parameters for ODMR dips at $B = 0$ G, using both Lorentzian and Generalized Gaussian functions for CW and pulsed protocols.

Parameter	CW-ODMR	Pulsed ODMR
<i>Lorentzian Fit</i>		
x_0	2870 ± 0.0026	2870 ± 0.32
γ	32.46 ± 0.0068	9.264 ± 0.52
A	25.23 ± 0.0062	1.765 ± 0.082
D	$1.024 \pm 2.1 \times 10^{-5}$	0.9962 ± 0.00054
<i>Generalized Gaussian Fit</i>		
x_0	2870 ± 0.049	2870 ± 0.3
γ	37.48 ± 0.13	12.14 ± 0.41
β	1.487 ± 0.011	2.646 ± 0.34
A	27.54 ± 0.17	2.021 ± 0.072
D	0.9999 ± 0.00048	0.9936 ± 0.00044

Source: By the author.

- A is a positive amplitude parameter that controls the *depth* of the dip; the negative sign reflects that the fitted feature is a minimum (inverted peak, *i.e.*, a dip).
- x_0 is the center of the dip, indicating the position of maximum signal attenuation.
- γ is a width parameter: for the Lorentzian, it corresponds to the half-width at half-maximum (HWHM); for the generalized Gaussian, it plays a similar role as a scale parameter that governs the spread of the dip.

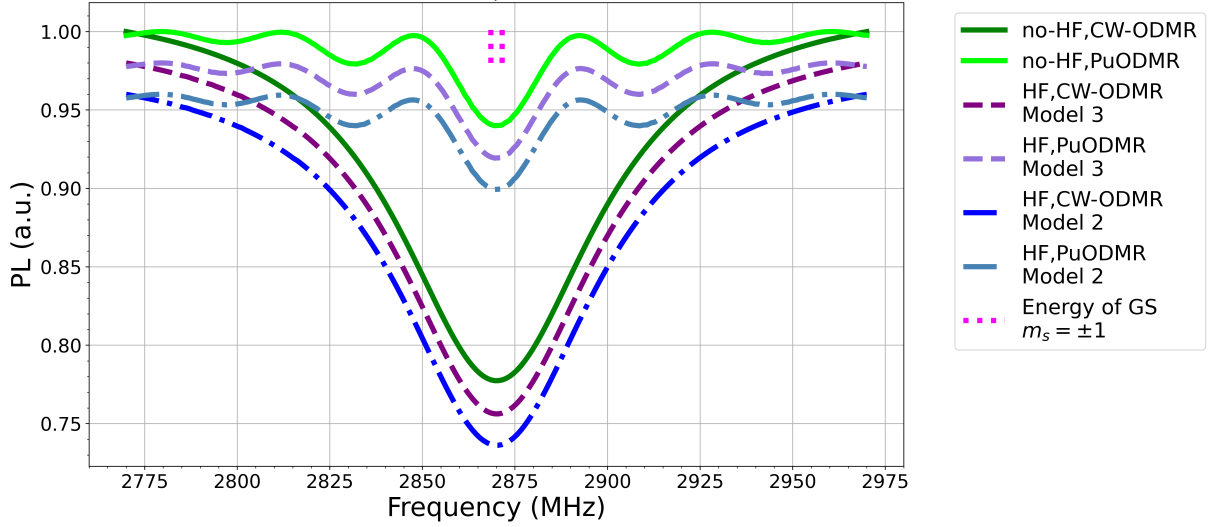


Figure 26 – Comparison between Pul-ODMR vs CW-ODMR ($\vec{B}_0 = 0$ G), in all we kept $W_I = W_R = 1.9$: CW-ODMR (darker lines) and Pul-ODMR (lighter lines), with models 2 (blue/bottom lines), 3 (purple/middle lines) and without hyperfine (green/top lines). We dislocated each dip of each situation by a little for them to not overlap too much, the overlaps happen only between CW-ODMR and Pul-ODMR in the same model.

Source: By the author.

Table 12 – Fit parameters for both ODMR dips at $B = 150$ G, using Lorentzian and Generalized Gaussian functions for CW and pulsed protocols.

Parameter	CW-ODMR		Pulsed ODMR	
	Dip 1	Dip 2	Dip 1	Dip 2
<i>Lorentzian Fit</i>				
x_0	2450 ± 0.013	3290 ± 0.0094	2450 ± 0.18	3290 ± 0.19
γ	25.02 ± 0.028	24.99 ± 0.021	12.52 ± 0.41	12.52 ± 0.41
A	16.42 ± 0.019	16.46 ± 0.014	3.584 ± 0.10	3.592 ± 0.10
D	1.0010 ± 0.000055	1.0010 ± 0.000041	1.0010 ± 0.00062	1.0010 ± 0.00062
<i>Generalized Gaussian Fit</i>				
x_0	2450 ± 0.16	3290 ± 0.16	2450 ± 0.11	3290 ± 0.11
γ	32.69 ± 0.45	32.73 ± 0.44	14.74 ± 0.22	14.75 ± 0.22
β	1.362 ± 0.032	1.363 ± 0.031	2.269 ± 0.096	2.268 ± 0.096
A	22.10 ± 0.35	22.22 ± 0.34	3.908 ± 0.068	3.916 ± 0.069
D	0.9941 ± 0.00072	0.9950 ± 0.00069	0.9982 ± 0.00032	0.9982 ± 0.00032

Source: By the author.

- D is a vertical offset that represents the baseline level of the signal.

Each function includes a normalization factor that ensures unit area when $A = 1$:

- In the Lorentzian, the factor $\frac{1}{\pi\gamma}$ normalizes the total area under the curve to one.

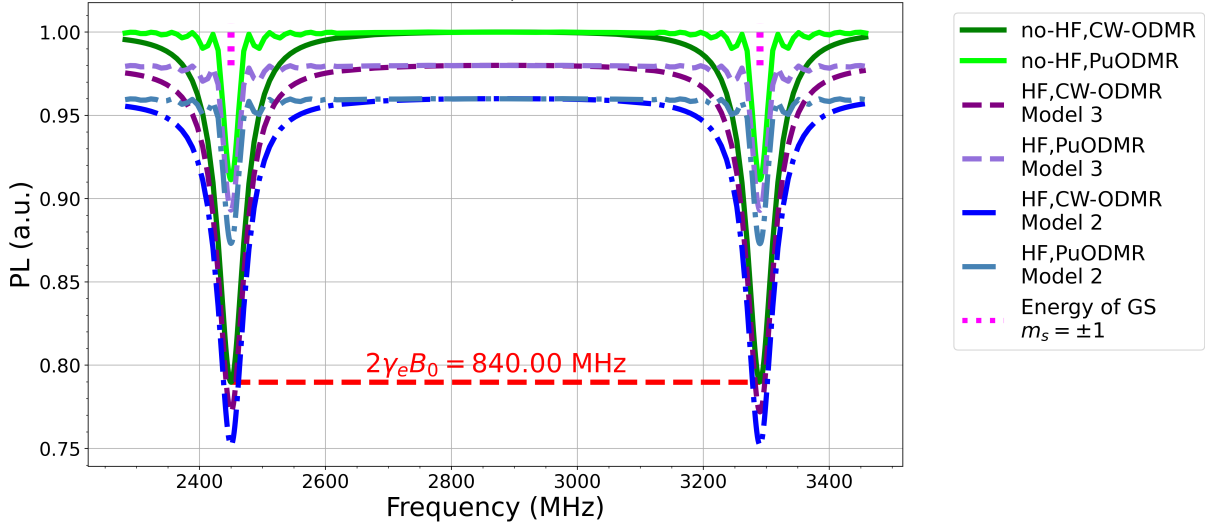


Figure 27 – Comparison between Pul-ODMR vs CW-ODMR ($\vec{B}_0 = 150$ G), in all we kept $W_I = W_R = 1.9$: CW-ODMR (darker lines) and Pul-ODMR (lighter lines), with models 2 (blue/bottom lines), 3 (purple/middle lines) and without hyperfine (green/top lines). We dislocated each dip of each situation by a little for them to not overlap too much, the overlaps happen only between CW-ODMR and Pul-ODMR in the same model.

Source: By the author.

Table 13 – Fit parameters for both ODMR dips at $B = 510$ G, using Lorentzian and Generalized Gaussian functions for CW and pulsed protocols.

Parameter	CW-ODMR		Pulsed ODMR	
	Dip 1	Dip 2	Dip 1	Dip 2
<i>Lorentzian Fit</i>				
x_0	1442.0 ± 0.05	4298 ± 0.04	1441.0 ± 0.2	4299.0 ± 0.2
γ	24.4 ± 0.1	24.81 ± 0.08	12.2 ± 0.3	12.2 ± 0.31
A	14.68 ± 0.06	15.37 ± 0.05	3.21 ± 0.07	3.36 ± 0.07
D	0.9999 ± 0.0001	1.0000 ± 0.0001	1.0000 ± 0.0003	1.0000 ± 0.0003
<i>Generalized Gaussian Fit</i>				
x_0	1442.0 ± 0.1	4298.0 ± 0.1	1441.00 ± 0.09	4299.00 ± 0.09
γ	32.7 ± 0.4	33.2 ± 0.4	14.8 ± 0.2	14.8 ± 0.2
β	1.31 ± 0.03	1.30 ± 0.03	2.25 ± 0.08	2.25 ± 0.08
A	20.8 ± 0.2	21.8 ± 0.3	3.65 ± 0.05	3.82 ± 0.05
D	0.9965 ± 0.0004	0.9967 ± 0.0004	0.9990 ± 0.0002	0.9990 ± 0.0001

Source: By the author.

- In the generalized Gaussian, the factor $\frac{\beta}{2\gamma\sqrt{\pi}\Gamma(\frac{1}{\beta})}$ serves the same purpose, ensuring the area is unity for all $\beta > 0$, here $\Gamma(\cdot)$ is Euler's gamma function.

In the generalized Gaussian, the parameter β controls the shape of the dip:

- $\beta = 2$ recovers the standard Gaussian profile;

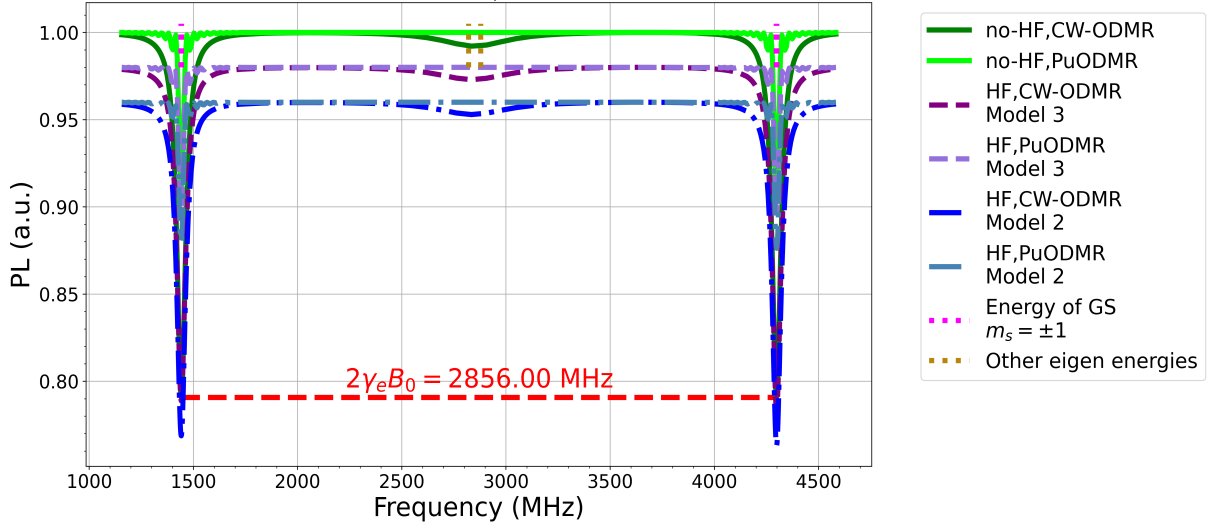


Figure 28 – Comparison between Pul-ODMR vs CW-ODMR ($\vec{B}_0 = 510$ G), in all we kept $W_I = W_R = 1.9$: CW-ODMR (darker lines) and Pul-ODMR (lighter lines), with models 2 (blue/bottom lines), 3 (purple/middle lines) and without hyperfine (green/top lines). We dislocated each dip of each situation by a little for them to not overlap too much, the overlaps happen only between CW-ODMR and Pul-ODMR in the same model.

Source: By the author.

- $\beta = 1$ yields the Laplace (double exponential) distribution;
- Smaller values of β lead to sharper dips with heavier tails.

We fitted both models to the data from the simulations and with the resulting fitting parameters we expected to recover the values that were set in the simulations for D_{GS} when $B_0 = 0$ and the values for B_0 retrieved from $x_{0,d_2} - x_{0,d_1} = 2\gamma_e B_0$, where these values x_{0,d_i} indicate the center of each dip. The values of the fitted parameters for each choice of B_0 are in Tables 11, 12, and 13. We can see the comparison between Pul-ODMR and CW-ODMR in Figs. 26, 27 and 28, where in the last we see that in CW-ODMR it is possible to detect a small dip near ES state resonance a feature not present in Pul-ODMR.

4.4 Results: Run times

In Fig. 29, it is presented how the addition of HF interaction of ^{15}N affects the simulation run time and how parallelization of simulations affects the simulation run time. We have run a sequence consisting of Laser, MW, and free evolution 10 times, and done the same 10 runs in a parallel way using the `loky_map` method in `QuTiP`. The plot shows the comparison of the total time of 10 sequential runs of the same sequence between repeated 5 times for having an average of the total time and the same thin, but for running the the 10 sequences in parallel, this was run for 3 setups using models 2 and 3, and doing

a run without any hyperfine term. The values of these are presented in Table 14. Fig. 30 does the same, but for 8 executions of the sequence, the data is in Table 15.

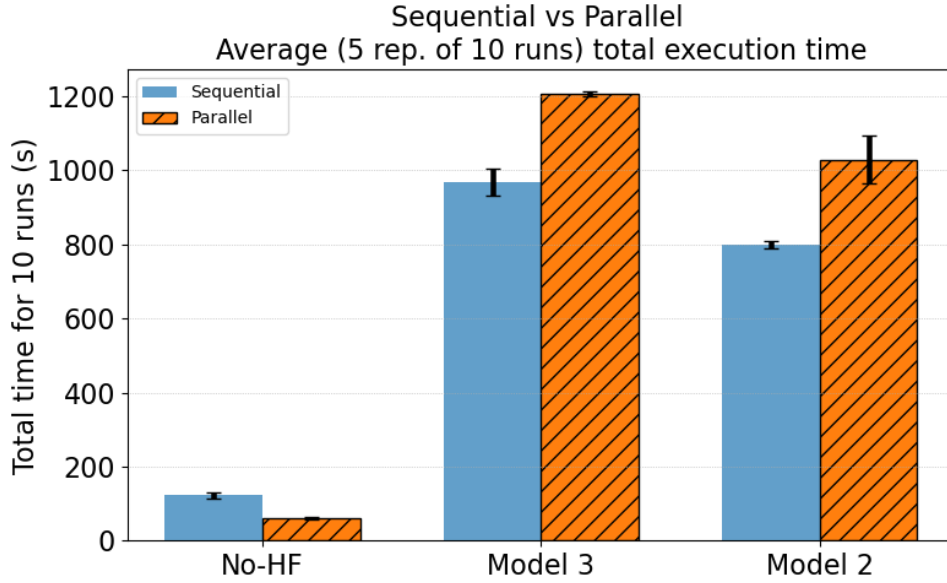


Figure 29 – Run time comparison average of 10 executions of a sequence involving Laser, MW and free evolution, between no-HF and HF execution with models 2 and 3 and parallel(orange/hash) vs non-parallel (blue/plain) execution using QuTiP with parallelization done via `loky_map` method in QuTiP

Source: By the author.

Table 14 – Average total time (10 seq.) over 5 runs (in s)

	Sequential	Parallel
No-HF	122 ± 7	60 ± 3
Model 3	967 ± 36	1206 ± 7
Model 2	800 ± 11	1030 ± 65

Source: By the author.

Table 15 – Average total time (8 seq.) over 5 runs (in s)

	Sequential	Parallel
No-HF	108 ± 8	47 ± 1
Model 3	811 ± 87	1050 ± 47
Model 2	653 ± 9	1051 ± 139

Source: By the author.

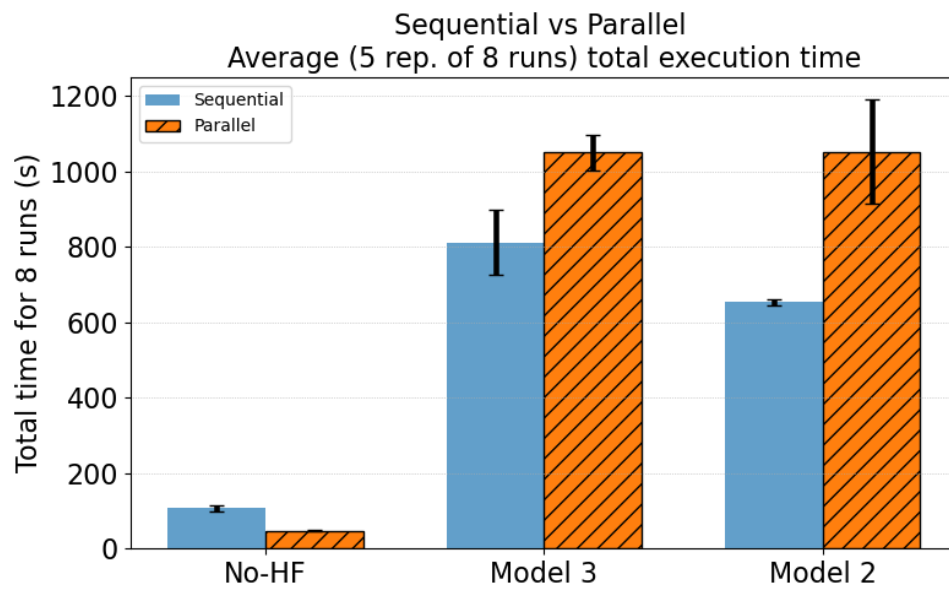


Figure 30 – Run time comparison average of 8 executions of a sequence involving Laser, MW and free evolution, between no-HF and HF execution with models 2 and 3 and parallel(orange/hash) vs non-parallel (blue/plain) execution using QuTiP with parallelization done via `loky_map` method in QuTiP

Source: By the author.

5 CONCLUSIONS AND PERSPECTIVES

This work investigated how different modeling choices for the NV center, particularly for the hyperfine (HF) interaction, affect the simulated dynamics and readout protocols. We observed, for example, that naive approaches (*ad hoc* insertion of HF terms in a simplified Hamiltonian), such as the one shown by Fig. 14, may result in unrealistic or misleading behaviors. In contrast, treating the system with more rigor, as in Models 2 and 3, reproduces the expected physical processes and leads to mutually consistent predictions. The population-dynamics results in Figs. 15, 16, 17 and 18 show that: (i) for sufficiently strong microwave driving, quantified by Ω_R , the differences between HF models are suppressed (Fig. 15); (ii) HF coupling becomes relevant near level anticrossings, producing detectable changes around ESLAC (Fig. 17); and (iii) the absence of a static magnetic field alters the MW dynamics and enhances the sensitivity to modeling details (Fig. 16). These comparisons clarify when HF terms are essential and when simplified models can still be used safely.

Another aspect explored was how realistic state preparation via optical pumping differs from assuming ideal pure states, as illustrated in Fig. 19. Prepared states yield PL profiles that differ systematically from those obtained under pure-state initialization; nevertheless, for $B_0 = 100$ G the presence or absence of HF coupling has only small effects on the PL. The PL profiles predicted by Models 2 and 3 also do not show visible differences in this regime (Fig. 22). Even so, the PL response can change with the microwave strength: varying Ω_R modifies the contrast and, when no magnetic field is present, can invert which prepared state is identified from PL for sufficiently strong driving (Figs. 23 and 24). When a static field is applied, Ω_R mainly changes the PL contrast between $|m_S = 0\rangle$ and $|m_S = \pm 1\rangle$. We also observed how PL changes depending on W_P (Fig. 21) and that it reproduces well the qualitative trends reported in the literature (Fig. 5). Our simulations also replicate the steady-state PL reduction under field misalignment, which depends on the orientation of $\vec{B}_0$ relative to the NV quantization axis, as shown by comparing the literature result in Fig. 7 with our simulations in Fig. 25.

In the ODMR spectrum of both protocols (continuous and pulsed), we observed that the zero-field splitting and the ground-state resonance frequencies can be recovered, with the additional benefit in CW-ODMR of obtaining a weak signature of an excited-state resonance. We also confirmed that the CW-ODMR dips are broader than the Pulsed-ODMR dips, as reported in Barry *et al.*⁵. We could not obtain clearly resolved resonances in the HF regime because, for the present dephasing and MW-power parameters, the hyperfine splittings remain unresolved within the simulated linewidth, being washed out by power and inhomogeneous broadening. As discussed in Ref. 5, the sensitivity in this

regime should improve if $\Omega_R \sim \pi/T_2^*$. For the value of $T_2^*(= 1.5\ \mu\text{s})$ chosen here, this corresponds to $\Omega_R \sim 2\text{ MHz}$, which is too small to lead to a full population transfer in the simulated MW oscillations. Therefore, using $\Omega_R \sim 16\text{ MHz}$, we were unable to systematically compare how the HF-resolved ODMR response evolves with Ω_R , leaving this as a direction for future work. In either case, we reproduced most of the expected ODMR behaviors.

Another difficulty observed was the performance of the simulation. While we achieved a low execution time for a single run or a few runs of varied sequences, we had a significant downside when obtaining data over large parameter sweeps. This was even more problematic when running the readout protocols, since they required multiple sequences for each point in the parameter space. We can see that running a model without an HF interaction took a reasonable amount of execution time, but that time increases quickly when running models with an HF interaction. Parallelizing the execution of protocols via `QuTiP`'s built-in methods reduced execution time only for the non-HF model. This behavior may change with GPU parallelization or more carefully designed parallel workflows. These observations are presented in Figs. 29 and 30 and Tables 14 and 15. We should note that runtime performance is platform-dependent and there are reports of significant differences between Windows/macOS and Linux implementations of CPU parallelization processing. We did not investigate this issue in our present study.

From the results gathered, we conclude that it is possible to effectively simulate NV centers using `QuTiP`, leveraging the tools implemented in this package to handle a wide range of experimental conditions. One example is to treat laser excitation as a bath of harmonic oscillators with the same frequency, as stated in Carmichael; Carmichael^{68, 69}, which can be turned on and off. It is also possible to include more than one nucleus at a time, with caution, since adding even a single HF interaction significantly increases the simulation runtime. An alternative is to improve parallelization routines, for instance, by using GPUs with `QuTiP-JAX` or by migrating the code to `QuantumToolbox.jl` (the Julia version of `QuTiP` maintained by the same organization) as these packages mature. From the simulations, we see that some effects are only observed due to hyperfine interactions and that simplified models can mislead results when new interactions are added to the system. Finally, preparing NV states via optical pumping does not necessarily yield perfect spin polarization: under our test conditions the polarization saturates around 80%, depending on the laser power and timing used, consistent with the literature.¹¹

Possible future steps for this work are: optimizing run time via parallelization and exploring how to reuse shared parts of state preparation across parameter sweeps. Identifying which parts of the state preparation are equal when varying control parameters, running them once, and branching after that to parallelize only the differing parts of the protocols is one possible strategy to save time. As another significant contribu-

tion from this work, we intend to turn the tools developed here into a modular part of the open-source package `QuaCCAToo`⁷⁰, for general use and further developments by the NV community.^I Finally, another interest is to create an easy way to use these tools in parameter-estimation pipelines for real NV systems that we would like to characterize or control using machine-learning techniques.⁷

^I For now, we have a baseline version of the code developed in this dissertation available on GitHub⁸. We will add a tutorial to replicate the main results, showing how to run the code. Later, we will also make it available on Zenodo with the full dataset generated by the code. The link to Zenodo will be available in the GitHub repository when we finish.

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